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"CGLOPS-2"

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ALGORITHM THEORETICAL BASIS DOCUMENT

LAKE WATERS

100M PRODUCT

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List of Acronyms

ATBD	Algorithm Theoretical Basis Document
BC	Brockmann Consult
CCI	Climate Change Initiative
C-GLOPS	Copernicus Global Land Operations
FTP	File Transfer Protocol
GLWD	Global Lakes and Wetlands Database
GPT	Graph Processing Tool
LSR	Lake Surface Reflectance
MERIS	Medium Resolution Imaging Spectrometer on Envisat
MSI	Multi Spectral Instrument on Sentinel-2
NERC	National Environment Research Council (UK)
NIR	Near-Infrared, part of the electromagnetic spectrum
obs	Observation
OC	Ocean Colour
OLCI	Ocean and Land Colour Instrument on Sentinel-3
OWT	Optical Water Type
PML	Plymouth Marine Laboratory
PUM	Product User Manual
QAR	Quality Assessment Report
rep	representative
R_{rs}	Above-water remote-sensing reflectance (units sr^{-1})
R_w	Fully normalised water-leaving reflectance
SWIR	Shortwave Infrared, part of the electromagnetic spectrum
TS	Trophic State
TSI	Trophic State Index
TSM	Total Suspended Matter
TUR	Turbidity
WGS84	World Geodetic System 1984

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1 BACKGROUND OF THE DOCUMENT

1.1 EXECUTIVE SUMMARY

The Copernicus Land Monitoring Service (CLMS) Lake Water Quality 100m (LWQ-100m) product provides an optical characterisation of inland waterbodies at 100m spatial resolution based on Sentinel-2 Multispectral Imager (MSI) data. This product was first provided between December 2018 and March 2020 (demonstration mode v1.3), to complement the LWQ-300m product that includes >4000 lakes globally and uses radiometrically and spectrally superior ocean colour instruments, but at lower spatial resolution. The 100m service has been continually active since July 2021 (v1.5) and has since been updated to the current product version (v2.0, v2.1), which aligns the 100m outputs with the 300m product. It continues to be provided in near-real-time.

Products are available for all resolvable water bodies within the outline of 225 Sentinel-2 MSI tiles over Europe and Africa. The 100m products contain six (sets) of variables: lake water-leaving reflectance (all wavebands available after atmospheric correction), turbidity, a trophic state index (derived from phytoplankton biomass by proxy of chlorophyll-*a*), total suspended matter concentration, chlorophyll-a concentration and a floating cyanobacteria risk index. Production and delivery cover up to 10-day intervals (starting the 1st, 11th and 21st day of each month), mapped onto a common global grid.

The current version (v2.1) is an update to the previous major release v2.0 (September 2024-October 2025). V2.1 contains a new scene classification layer based on the CLMS Water and Ice Cover (WIC) Sentinel-1+Sentinel-2 merged product. This new layer complements existing quality flagging in the LWQ-100m product, and provides an additional, independent source of lake ice/snow masking to users.

While Sentinel-2 MSI was primarily designed for land applications, some properties of waterbodies can also be observed, as demonstrated by the LWQ-100m product collection. The algorithms used to derive the input for the optical lake water products are implemented in the *Calimnos* processing chain, which also handles production of the Sentinel-3 OLCI LWQ-300m products. This Algorithm Theoretical Basis Document (ATBD) provides an overview of the applied algorithms, including the latest developments in the current product version.

1.2 SCOPE AND OBJECTIVES

One of the main objectives of the Copernicus programme is to provide appropriate satellite-derived information to the scientific community, member states and policy makers, for a range of applications. The Sentinel MSI based 100-m lake water quality products are currently delivered as demonstration products. The products demonstrate the feasibility and quality of Sentinel-2 MSI for water quality applications, where a higher resolution of the product is required than can be observed at up-to-daily intervals at 300m from current ocean colour sensors. The objective of this

document is to provide details on the algorithms and processing chain used for the optical LWQ-100m products.

1.3 CONTENT OF THE DOCUMENT

This document is structured as follows:

- Chapter 2 recalls the user requirements and the expected performance
- Chapter 3.2 describes the processing chain and applied algorithms of the current version
- Chapter 3.3 provides results of the quality assessment including current limitations
- Chapter 3.4 summarises the risk of failure and mitigation measures

1.4 RELATED DOCUMENTS

1.4.1 Applicable documents

AD1: Annex I – Technical Specifications JRC/IPR/2015/H.5/0026/OC to Contract Notice 2015/S 151-277962 of 7th August 2015

AD2: Appendix 1 – Copernicus Global land Component Product and Service Detailed Technical requirements to Technical Annex to Contract Notice 2015/S 151-277962 of 7th August 2015

AD3: Call for tenders JRC/IPR/2023/CPN/2325 – Framework Service Contract – Operation of the bio-geophysical variables systematic monitoring of the Global Land Component of the Copernicus Land Service 'CGLOPS' Part 2: Technical Specifications, Ref. Ares(2023)6699927, dated 03/10/2023

1.4.2 Input

Document ID	Descriptor
CGLOPS2_SSD	Service Specifications of the Global Component of the Copernicus Land Service
CGLOPS2_QAR_LWQ100_S2_v2.1.0_I1.00	Quality Assessment Report for 100m lake water quality demonstration products
CGLOPS2_PUM_LWQ100_S2_v2.1.0_I1.02	Product User Manual summarizing all information about the 100m lake water quality demonstration products
CGLOPS2_ATBD_Polymer_v1.0.0_I1.02	Description of Polymer Algorithm for different instruments

2 REVIEW OF USER REQUIREMENTS

According to applicable document [AD3], the user requirements relevant for Lake Water Quality products are:

- **Definition**:

The Lake Water Quality products v2.1 are lake turbidity (TUR) in NTU, total suspended matter (TSM) in g/m^3, Trophic State Index (TSI), chlorophyll-a concentration (Chl-a) in g/m^3 and lake surface reflectance spectra (LSR, given in terms of fully normalized water-leaving reflectance R_w; a dimensionless quantity) from which TUR, TSM, Chl-a, TSI and the floating cyanobacteria index are derived. The products shall be provided as 10-day averages of the respective parameters, except for LSR for which the observation that is statistically most representative for the observed time period is given. The reason for the latter is that an average of LSR is not physically accurate and would not be useful to feed into alternative algorithms for the retrieval of optical water quality parameters. LSR describes the apparent colour of the water body. Turbidity and TSM are key indicators of water clarity. Turbidity quantifies the haziness of the water and acts as an indicator of underwater light availability. TSM quantifies the concentration of suspended particles in the water body. TSI, as defined by Carlson (1977), refers to the degree at which organic matter accumulates in the water body and is most commonly used in relation to eutrophication monitoring. Chl-a is the precursor product to Trophic State Index and is commonly used as a proxy to phytoplankton biomass. The cyanobacteria risk index indicates the detection of cyanobacteria surface blooms in the water body.

Geometric properties:

The baseline datasets described in this document shall be provided at a resolution of 100m. The target baseline location accuracy shall be 1/3 of the at-nadir instantaneous field of view. Pixel coordinates shall be given for the pixel centre.

- **Geographical coverage**:

Global window

The initial window definition is aligned to the global datasets produced during the GMES Initial Operations (GIO) phase for the most widely used output data:

- geographic projection: lat lon,
- geodetical datum: WGS84
- accuracy: min 10 digits
- coordinate position: pixel centre
- global window coordinates: UL: 180W-75N, BR: 180E, 56S

The following output specifications are further optimised with respect to the requirements:

- pixel size at 100m: 0.1/112°
- global window coordinates: UL: 180°W-90°N, BR: 180°E, 90°S
- global grid size at 100m: 400752 columns, 200376 lines

Continental windows:

Continental windows may be asked by the contracting authority to satisfy specific user needs. Wherever applicable, continental windows shall be drawn from global output data.

- **Accuracy requirements**:

Lake Water-Leaving Reflectance is an Essential Climate Variable (GCOS 244, 2022) with an associated accuracy requirement threshold of 30% (breakthrough: 20% and target: 10%) at the reflectance peak (GCOS 245, 2022). Turbidity has an uncertainty threshold of 50% (goal: 35%) and trophic state has an uncertainty threshold of 0.3 in log10 or 50% [AD3].

- **Temporal Definition**

As a baseline, the biophysical parameters are computed by and representative of *dekads*, i.e. periods of nominally ten and varying from eight to eleven days, defined to start on days 1, 11 and 21 of each month of the year.

3 METHODOLOGY DESCRIPTION

3.1 OVERVIEW

CGLOPS Lake Water processing uses the *Calimnos* processing chain. *Calimnos* was initially developed in UK NERC GloboLakes (2012-2018) with input from the ESA Diversity-II project (2015) and has since been further developed with core collaboration between Plymouth Marine Laboratory (PML), University of Stirling, Brockmann Consult (BC) and HYGEOS in the Copernicus Land Monitoring Service, H2020 TAPAS, H2020 EOMORES, and ESA Lakes_cci activities. The chain can be used to process archived Envisat MERIS, Aqua-MODIS, Sentinel-3 OLCI and Sentinel-2 MSI data from archives or in near real-time. This document refers to 100m products based on MSI generated using *Calimnos v2.1*.

Calimnos combines into a single processing chain: data discovery, spatial resampling and subsetting by target area (individual water bodies, image granule or any other region of interest), radiometric and atmospheric corrections, pixel identification (land/cloud/water/ice), optical water type classification, individual algorithms (per parameter and water type), weighted blending of results, optional reprojection and spatiotemporal aggregation (Figure 1). For the production of 100m products *Calimnos* is operated on the Copernicus Data Space Ecosystem (CDSE).

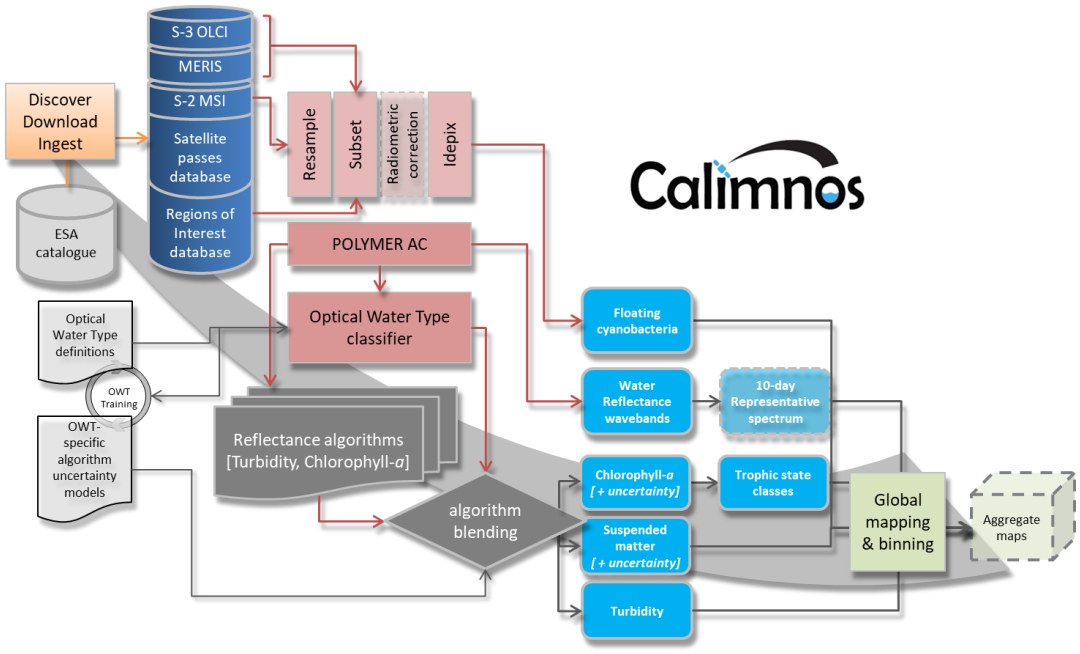


Figure 1. Schematic overview of the Calimnos processing chain v2.1, including MSI data processing

The following section provides an overview of the algorithms selected for the *Calimnos* processing chain for 100m products.

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3.2 ALGORITHMS

3.2.1 Outline

A schematic overview of *Calimnos* is given in Figure 1. The main processing stages include:

- Data discovery.

Granules matching regions-of-interest are identified and downloaded as they become available. Target regions are similarly specified in a geospatial database and satellite products which overlap any of the target regions can be queued for processing.

Tools: in-house Python scripts and PostgreSQL database functionality.

- Resampling.

The MSI bands at 10, 20, and 60m resolution are resampled to 100m. The Sentinel-2 resampling routine is part of the SNAP toolbox, called through the Graph Processing Tool (GPT).

Tools:

- SNAP S2-resampling routine; retrieved resolution is 100m
- SNAP Subset operator

- Subsetting.

This is an optional step and currently not applied.

Matching satellite passes can be subset to the area of interest. Currently, whole granules (UTM tiles) are processed.

Tools: The subsetting routine is part of the SNAP toolbox, called through GPT.

- Radiometric corrections.

This is an optional step which applies any radiometric corrections defined following the launch of the sensor to the L1C imagery before submitting the data to atmospheric correction. Lack of L1B (unmapped) imagery from Sentinel-2 is understood to prevent optimal image correction in the overlap between detectors, which cannot be corrected in this stage. Radiometric corrections are currently not applied to MSI.

- Pixel identification.

Pixel identification and classification over water with the *IdePix* algorithm for Sentinel-2. The algorithms combine several spectral tests and are applied for initial pixel identification as water, land, mixed pixels, cloud/haze, cloud shadow or snow/ice. Pixel identification masks are stored for later masking of invalid (non-water) pixels.

Tools: IdePix-Sentinel-2 v8.0.1 is called through SNAP using the GPT.

- Atmospheric correction.

Polymer is applied to the corrected L1C data that match the water mask derived from *IdePix* and yields water-leaving reflectance wavebands. Polymer is called using a wrapper function in Python.

Tool: Polymer v4.14 according to Steinmetz et al. (2011), updated for Sentinel-2 application, parameterised to use the Park and Ruddick (2005) bidirectional reflectance distribution function, including mineral absorption and initialization of the optical model with chlorophyll-a = 10 mg m^{-3} and suspended matter = 10 g m^{-3} for case-2 waters, and operating only on pixels identified as water by the *IdePix* module.

Polymer v4.15+ is the next candidate processor used for the Lake Water Quality products and includes a wider range of initialization conditions and range of backscattering albedo in the bio-optical model, compared to previous versions. The QAR document provides details on the performance of LWQ product v4.15 compared to v4.14. Product v2.0+ remains with the same atmospheric correction processor (Polymer v4.14), because downstream algorithms will need to be recalibrated before both can be included in the service, which is the current outlook towards LWQ-100m product v3.0.

- Optical water type classification.

Optical water type (OWT) classification derived from atmospherically corrected Sentinel-2 MSI imagery is applied to the image. Each pixel was initially assigned a membership score for four OWT classes. The classifier has been updated by including a wider range of optically diverse water bodies. With product version 2.0+, the OWT separates between 10 optical water classes.

Tool: Python at PML based on the work of Moore et al. (2001) and equivalent software developed for ESA Lakes_cci. The algorithm uses the spectral angle to determine similarity with predefined optical water types.

- Algorithm mapping and blending.

For each OWT, algorithms designed to be best suitable to describe chlorophyll-a and turbidity of that water type are blended on a pixel-by-pixel basis. A weighted average of turbidity and TSI (derived from Chl-a) is then produced based on the membership functions.

For OWTs where no algorithm performed well during validation no algorithm is assigned. Pixels with these OWTs are set to NaN.Algorithms CHL:

- OWT class 1, 5: OC2 oceancolor.gsfc.nasa.gov/cms/atbd/chlor_a
- OWT class 7: 708/665 empirical band ratio based on Gilerson et al. (2010)
- OWT class 2,9: Semi-analytical NIR-Red band algorithm for MERIS based on Gons et al. 2005.
- OWT class 3, 4, 6, 8, 10: unassigned.

Algorithms TUR: Nechad et al. (2010, 2016)

- OWT class 1, 2: signal waveband 665nm
- OWT class 5: signal waveband 705nm
- OWT class 9: signal waveband 783nm
- OWT class 7,10: signal waveband 865nm
- OWT class 3, 4, 6, 8: unassigned

- Floating cyanobacteria / macroalgae flag

Floating Algae Index (FAI) adopted from Dogliotti et al. 2018 and Oyama et al. 2015 to detect floating material and subsequent test based on Normalised Difference Water Index (NDWI) to distinguish between floating cyanobacteria and floating macroalgae.

- Temporal aggregation.

10-day aggregation is done on a per-region basis after applying masks to select valid data for water pixels. The reflectance products of the Polymer atmospheric correction are inspected over the 10-day period and for each pixel the most representative spectrum is chosen (see Section 4). The aggregated products which contain the reflectance, chlorophyll concentration, trophic state index, total suspended matter and turbidity parameters are then reprojected to a global grid.

Tools:

- SNAP Level-3 binner (LakeAggregate operator v1.3)
- Most representative spectrum selection. Algorithm developed for CGLOPS between PML and BC. For each pixel, water-leaving reflectance spectra are compared to the median spectrum using the root-mean-square (RMS) difference across all wavebands. The spectrum with the lowest RMS difference is selected as most representative for the observation period.
- Mosaicking.

The 10-day aggregated products are merged into a global grid using a Python script configured to select the pixel with the least spatial displacement in the region of overlap between MSI granules from the same orbit (Roy et al. 2016). This process is facilitated through a look-up table providing the pixel displacement for each pixel in the MSI UTM tiles compared to the 100-m global output grid, which is considered constant between orbits.

The SNAP version used is 8.8cv.

3.2.2 Basic underlying assumptions

The processing chain is based on a number of assumptions:

- Sentinel-2 MSI has sufficient radiometric sensitivity and dynamic range to capture water leaving reflectance in each band at 100m resolution, in dark, clear, and turbid waters.
- *IdePix* can differentiate adequately between pixels containing water and any of the following conditions: mixed land/water, cloud, cloud shadow, ice, snow, and haze.
- *Polymer* successfully retains the shape and amplitude of water-leaving reflectance.
- The optical water types derived from analysing MSI imagery over a diverse set of water bodies sufficiently capture the diversity of natural water types observed in CGLOPS.
- Algorithms calibrated to the narrow wavebands of MERIS and OLCI are applicable to the nearest located MSI wavebands, such that calibration of these algorithms yields results that are comparable with those from medium resolution imaging sensors.

- In case multiple algorithms for Chl-a/TSI and turbidity are used for different optical water types, it is assumed that the water-leaving reflectance spectra can be assigned a membership score for at least one known optical water type.
- The (tuned) reflectance algorithms for chlorophyll-*a* and suspended matter yield valid results over the range of reflectance values observed over water types to which they have been assigned.
- The translation of chlorophyll-*a* to trophic state is adequate and consistent between water types.
- Selection of the most representative spectrum from observations obtained over 10 days uses statistical metrics that result in a consistent pattern for the whole water body.

A crucial assumption in each version of the *Calimnos* processing chain is that predefined optical water types contain sufficient similarity to each satellite observation (pixel) containing open water. Figure 2 shows the ten types that are used in the current version for MSI.

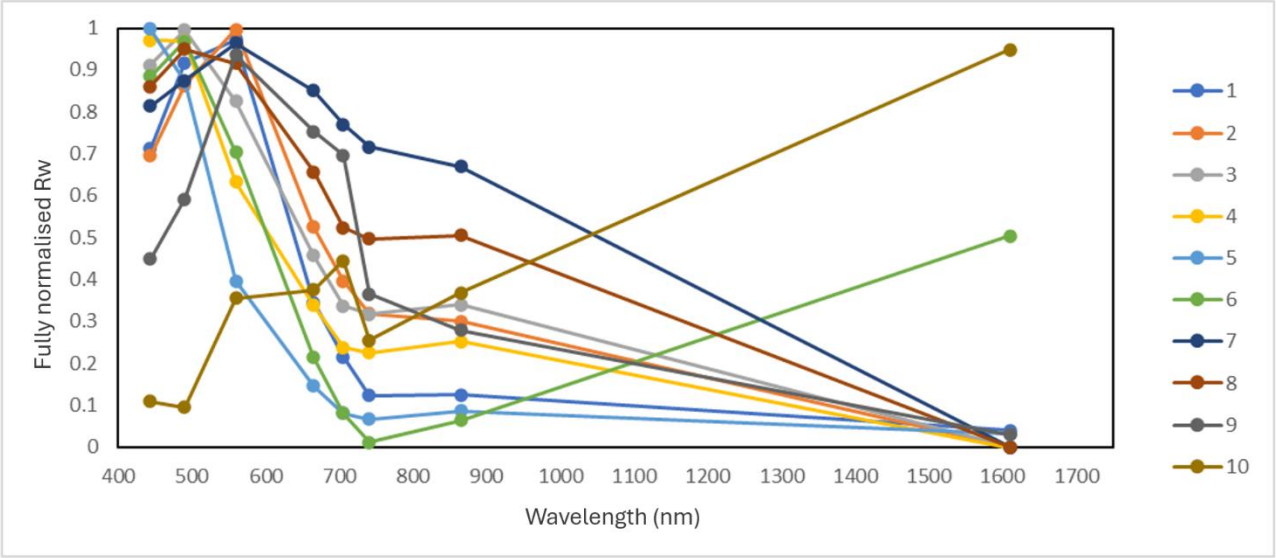


Figure 2. Optical water type spectra for Sentinel-2 MSI

In practice, the assignment to an optical water type can fail due to a number of reasons, the most prominent being failure to identify the pixel as water (or as influenced by cloud or ice) by *IdePix*, or failure in the atmospheric correction leading to a low similarity to known optical water types. Optical water type membership scores are always assigned; poor similarity to any known type may result in downstream algorithms producing inconsistent results that require subsequent masking or flagging, which is done in case of extreme, or negative results.

3.2.3 Related and previous applications

The first CGLOPS release used *Calimnos* v1.1 for reprocessing of the MERIS archive (see LWQ-300m product). *Calimnos* v1.04 was used to produce the first global, adaptive water quality dataset in the UK NERC GloboLakes project to relate water quality changes to environmental change in

1000 lakes globally. Compared to the GloboLakes project processing, the binning procedure has been changed to retrieve Lake Surface Reflectance of the most representative pixel, and to aggregate the Trophic State Index from the chlorophyll-a concentration. The aggregation period is set to 10 days. The selection of lakes in the current CGLOPS production is additionally based on different criteria (large and strategic water bodies) compared to GloboLakes. The CGLOPS selection of 225 MSI tiles largely overlaps with the European and African waterbodies included in the 300m resolution service. The latter, with more than 4200 waterbodies included globally, covers approximately 70% of the global surface area of inland water bodies as defined in the Global Lakes and Wetlands Database (GLWD) Level 1 and 2. The current version for 100m (and 300m) NRT production is *Calimnos* v2.1.

Calimnos is also used to generate climate data records for the Lakes Essential Climate Variable as part of the ESA Climate Change Initiative (CCI), which currently includes product v3.0. Improvements to product quality and consistency developed in ESA Lakes_cci are typically and periodically adopted in the operational Copernicus Land Monitoring Service (CLMS), for example per-pixel uncertainty estimates, and algorithm calibrations. Upgrade of the CLMS products to v3.0 is anticipated. Some key differences between ESA CCI and CLMS outputs are:

- ESA Lakes_cci provides an integrated product of the full Lakes ECV (including lake water level, extent, skin temperature, ice cover and lake colour products), at daily intervals and 1/120 degree resolution. Consistency between the products is used to identify and mask non-physical observations, for example, high reflectance in the presence of thin ice. The ESA Lakes_cci lake water quality products are derived from Envisat MERIS and Sentinel-3 OLCI observations of 2000 lakes, and Terra MODIS observations for a smaller number of lakes. Datasets are released every few years.
- CLMS LWQ products are provided at close-to-sensor native resolution (100m and 300m) from Sentinel-2 MSI, Envisat MERIS and Sentinel-3 OLCI, at 10-day aggregation periods. All ESA Lakes_cci water bodies are also included in the operational CLMS service. As an operational service, consistency with other observation types (ice, temperature) is not part of quality control procedures. The data are delivered with short latency (3-4 days).

Users of the CLMS LWQ products and/or Lakes_cci Climate Research Data Package may further be interested in the upcoming addition of Lake Water Quality to the Copernicus Climate Change Service (C3S) which will see 6-monthly releases of product version 3.0, at 300-m daily resolution, including 2000 water bodies. The latency of the C3S product will be 6-12 months from acquisition after the initial ramp-up phase.

3.2.4 Alternative methodologies currently in use

Several algorithms are available for Sentinel-2 MSI dedicated to atmospheric correction and water constituent retrieval. Atmospheric correction is provided by default with the official ESA Tool

Sen2Cor, which is not specifically designed for atmospheric correction over water[1]. VITO developed the iCOR Atmospheric correction which is based on aerosol retrieval over land with extrapolation to the water areas (Sterckx et al. 2015). It can be coupled with the SIMEC adjacency correction (Sterckx et al. 2015). ACOLITE is a development of RBINS, optimised for turbid coastal waters (Vanhellemont & Ruddick, 2014) and under active development. The latter two algorithms are available as standalone tools, while iCOR is also available through the ESA SNAP toolbox. A coupled system for atmospheric correction and water retrieval is provided by the C2RCC processor available with the ESA SNAP toolbox. Results from a comprehensive analysis of the performance of these processors and Polymer v4.10 is provided in Warren et al. (2019). Polymer and C2RCC consistently outperformed other processors in coastal and inland waterbodies, and preference was given to Polymer (LWQ-100m products v2.0 and v2.1 using v4.14 and v4.17+ for v3.0) for consistency with the CLMS LWQ-300m service.

For *Calimnos* v1.5 (LWQ-100m v1.5), algorithms for chlorophyll-a and turbidity were tuned against S-3 OLCI observations acquired on the same day, as detailed in Warren et al. (2021). The requirement for algorithm tuning implies that with every significant change in atmospheric correction, downstream algorithms also require recalibration and subsequent uncertainty characterisation. This process will need to be repeated when, for example, Polymer 4.17+ is adopted in LWQ product version v3.0.

Several new methods have emerged in recent years for the estimation of chlorophyll-a and other biogeochemically relevant variables. The use of Mixture Density Networks (Pahlevan et al. 2020, Smith et al. 2021, Pahlevan et al. 2022) is of interest and bears some similarity to data-driven training operations as those deployed in the service via OWT-specific calibration of previously published algorithm formulations. Another alternative uses Bayesian Neural Networks (Werther et al. 2022) and was shown to improve performance in oligotrophic and mesotrophic water types. Both approaches have been applied to MSI and OLCI data by their developers, and they are included in ongoing round-robin evaluations in the ESA CCI Lake project, feeding results and expert knowledge into CGLOPS, when applicable.

3.2.5 Input data

Satellite input data:

- Sentinel-2 MSI L1C, downloaded from ESA open data hub.

Digital Elevation Model (*IdePix*, Polymer):

- For regions with a maximum latitude < 60 degrees, the SRTM 1Sec HGT is used. For regions that cross or are over 60 degrees latitude, GETASSE30 is used.

Polymer ancillary data:

[1] ESA Sen2Water products are atmospherically corrected for inland and coastal water surfaces, corresponding to the current LWQ100 product. Public release of Sen2Water products is expected by 2027.

- GMAO_FP NASA ancillary files.

Optical water type definitions (means and covariance matrices):

- A specific set of Optical Water Types for MSI (Figure 2).

3.2.6 Output product

Sentinel-2 MSI granules, optionally subset to regions of interest (water bodies) have been processed to NetCDF4 format compliant with CF convention 1.8. The data are mapped on a 100m global grid.

Each file includes the bands described in Table 1. Once mapping to the global grid takes place, the band name specifications may change slightly to represent data aggregation methods (e.g. 'turbidity' becomes 'turbidity_mean').

Table 1. Variables (bands) provided in the output product

Band name	Data type	Description
num_obs	int	Number of observations of the pixel area (including masked and invalid observations)
first_obs	float	Age in days after the start of the observation period of the first observation included in the pixel area
last_obs	float	Age in days after the start of the observation period of the last observation included in the pixel area
turbidity_mean	float	Mean of turbidity in Nephelometric Turbidity Units (NTU) (in the observation period)
tsm_mean	float	Mean of concentration of total suspended matter (in the observation period)
chla_mean	float	Mean concentration of chlorophyll-a (in the observation period)
trophic_state_index	float	Trophic state index (TSI) derived with *Calimnos* blending of chlorophyll-a algorithms tuned in the UK NERC GloboLakes project (averaged over the observation period. See Table 2 for a mapping of chl-a values to TSI)
R_w443_rep	float	Representative spectrum of fully normalised water-leaving reflectance at [the waveband indicated in the band name]
R_w490_rep	float	
R_w560_rep	float	
R_w665_rep	float	
R_w705_rep	float	
R_w740_rep	float	

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R_w783_rep	float	
R_w842_rep	float	
R_w865_rep	float	
R_w945_rep	float	
R_w1375_rep	float	
R_w1610_rep	float	
R_w2190_rep	float	
R_wDayNum_rep	float	Age in days after the start of the observation period of the most representative R_w spectrum
floating_cyanobacteria	float	Probability of floating cyanobacteria (with a range of values from 0 to 1 indicating probability of cyanobacteria presence – the closer the value is to 1, the higher the probability)
lat	float	Latitude in decimal degrees
lon	float	Longitude in decimal degrees
time	float	Time in days since 1-1-1970
crs	string	Geographical projection string
quality_flags	int	Flag band showing why input observations to a pixel have been included. Flags are L1_invalid, bright_pixels, cloud, snow_ice, land, turbidity_invalid, chla_invalid, tsm_invalid, atmospheric_correction_failure, floating_cyanobacteria, floating_macrophytes_or_mixed_pixel
WIC_S1S2_aggregate	int	Scene classification based on aggregated information from the CLMS Water and Ice Cover (WIC) Sentinel-1 and Sentinel-2 merged product (WIC S1+S2)

3.2.7 Algorithm Description

An overview of the individual modules and algorithms is given below, followed by detailed procedures for each processing stage:

Preprocessing:

- SNAP resampling to 100m resolution

Pixel identification:

- IdePix.S2 v8.0.3 Pixel identification and classification for identifying non-valid water pixels influenced by clouds, cirrus, cloud shadow or mixed pixels or land.

- From product version 2.1, snow/ice and other relevant masking information are additionally sourced from the CLMS Water and Ice Cover (WIC) Sentinel-1 and Sentinel-2 merged product.

Atmospheric correction:

- Polymer v4.14 according to Steinmetz et al. (2011), parameterised to use the Park and Ruddick (2005) bidirectional reflectance distribution function, and operating only on pixels identified as water by the *IdePix* module.

Optical Water Type (OWT) classification:

- Implemented in Python at PML based on the work of Moore et al. (2001) and equivalent software developed for UK NERC GloboLakes and ESA Lakes_cci. The algorithm uses the spectral angle to determine similarity with predefined optical water types.

Water constituent algorithm mapping:

- Water constituent retrieval algorithms considered suitable for specific OWTs are mapped to each individual pixel. Mapped algorithms are then averaged using the weights of the three highest OWT similarity scores to generate the blended turbidity or chlorophyll-a product. Trophic State Index is then derived from the chlorophyll-a concentration.

Floating Cyanobacteria / Macroalgae flag.

- Floating Algae Index (FAI) adopted from Dogliotti et al. 2018 and Oyama et al. 2015 to detect floating material and subsequent test based on NDWI to distinguish between floating cyanobacteria and floating macroalgae.

Temporal aggregation and reprojection (L3 processing)

- The SNAP LakeAggregate operator (v5.0.9) performs spatial and temporal aggregation of pixel values into cells ('bins') of a planetary grid over a lake. During the Lake Aggregation step, invalid pixels are masked before averaging. The standard deviation and mean values are calculated for the remaining observations over a pixel. Invalid pixels are recognised based on L1 product flags (L1.INVALID), Polymer masks (OUT_OF_BOUNDS) the *IdePix* cloud, ice, and land identification flags, and value limits for individual products (Water-leaving reflectance non-negative and not exceeding 0.4 for 443 and 865nm, turbidity and TSI non-negative) (see Table 5).
- Most representative spectrum selection. For each pixel, water-leaving reflectance spectra are compared to the median spectrum using the root-mean-square difference across all wavebands. The spectrum with the lowest RMS difference is selected as most representative for the observation period.

Mosaicking

- The 10-day aggregated products are merged into a global grid, selecting the pixel with the least spatial displacement in the region of overlap between MSI granules from the same orbit (Roy et al. 2016). This process is facilitated through a look-up table providing the pixel

displacement for each pixel in the MSI UTM tiles, compared to the 100m global output grid, which is considered constant between orbits.

3.2.7.1 Pre-processing

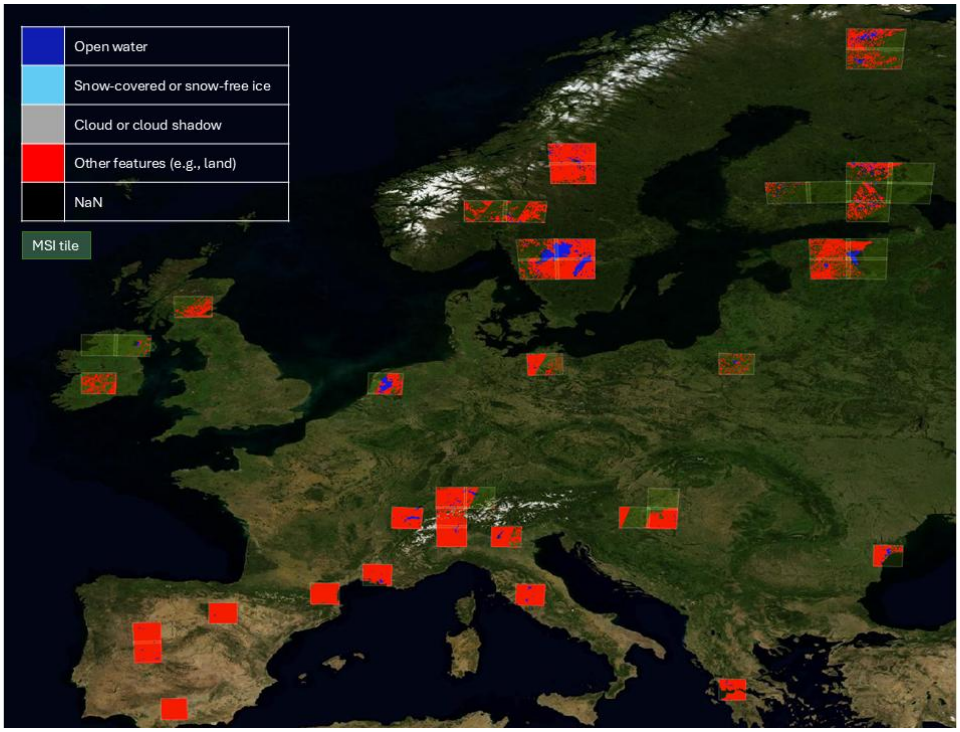
For the processing of MSI granule no additional pre-processing other than resampling is done or foreseen. Sentinel-2 MSI image data are provided by ESA at L1C (mapped). This projection is kept throughout the L2 processing stages.

3.2.7.2 Pixel Identification

Pixel identification is needed as pre-processing step to identify water pixels for downstream processing. This is performed by the IdePix.S2 v8.0.3 processor available in SNAP. It uses spectral tests for the identification of cloud, haze or land influenced pixels. The *IdePix* classification processor also includes cloud shadow identification. Primarily, the cloud shadow algorithm depends on a correct cloud mask for which the areas of potential cloud shadow based on the illumination geometry are identified. In one approach, a clustering of these areas provides the darkest pixels which are flagged as cloud shadow. In a second approach, the cloud mask is shifted along the illumination path until the minimum of the mean value in reflectance is reached. In a final cloud shadow flag, the results of both methods are combined.

In addition to the IdePix outputs, lake snow/ice and other relevant masking information (e.g., open water and cloud) sourced from the CLMS Water and Ice Cover (WIC) Sentinel-1 and Sentinel-2 merged product (WIC S1+S2) is now also contained as an additional, new layer in the LWQ-100m product v2.1. This product uses Sentinel-1 synthetic aperture radar and Sentinel-2 MSI and is generated on a daily basis for inland water within the EEA38+UK area defined by a dynamic water mask that is updated annually. It provides information about river and lake areas covered by snow-covered or snow-free ice at 20m spatial resolution (HR-WSI PUM 2025). No WIC information is provided for MSI tiles outside the EEA38+UK region.

An example LWQ-100m v2.1 product (‘WIC_S1S2_aggregate’ layer) is shown below, overlaid on NASA’s Blue Marble map; first dekad in July 2025 (period 01/07/2025-10/07/2025):



3.2.7.3 Atmospheric Correction (Lake Water Reflectance)

Polymer is an atmospheric correction processor originally developed for clear ocean (Case-1) water, and specifically able to deal with sunglint (Steinmetz et al. 2011). Recent improvements (up to v4.15) make the processor increasingly suitable to deal with a variety of optically complex (including inland) waters. Polymer applies a spectral optimisation based on a bio-optical model in conjunction with radiative transfer models to separate atmospheric (including sunglint) and water reflectance. The principle of the algorithm is a spectral matching method using a polynomial to model the spectral reflectance of the atmosphere and sunglint, and a bio-optical forward reflectance model for the water part. The algorithm uses the full set of wavebands available (user-configurable) as opposed to alternative ocean-colour methods that primarily extrapolate from near infra-red bands.

Polymer v4.15 or higher is the candidate processor used for the Lake Water Quality products and includes a wider range of initialisation conditions and range of backscattering albedo in the bio-optical model, compared to previous versions. The QAR document provides detail on the performance of LWQ product v4.15 compared to v4.14. LWQ-100m v2.0 and v2.1 remain with the same the atmospheric correction processor version (Polymer v4.14), because downstream algorithms will need to be recalibrated before both can be included in the service, which is the current outlook towards LWQ product v3.0.

3.2.7.4 Trophic State Index

Trophic State Index is obtained from chlorophyll-*a* according to Carlson (1977), as shown in Table 2.

Table 2. Classification of Trophic State Index with TSI and chlorophyll-a concentration scales

Trophic classification	Trophic State Index	Chlorophyll-a (mg/m^3)
Oligotrophic	0	0 - 0.04
	10	0.04 - 0.12
	20	0.12 - 0.34
	30	0.34 - 0.94
Mesotrophic	40	0.94 - 2.6
	50	2.6 - 6.4
Eutrophic	60	6.4 - 20
	70	20 - 56
Hypereutrophic	80	56 - 154
	90	154 - 427
	100	427 - 1183

The Trophic State Index is labelled with the upper boundary of the range of trophic state (TS) values it represents, e.g. a TSI of 10 represents 0 ≥ TS < 10.

TSI uses satellite-derived chlorophyll-*a* as input based on the following algorithms, which differ per Optical Water Type. In Table 3, the algorithm mapping applied in *Calimnos* v1.5 for MSI is shown. Five out of ten OWTs are mapped to these algorithms, based on the strength of the observed correlation between MSI and OLCI results for these observations as shown by Warren et al. (2021). Consequently, observations with high spectral similarity to the remaining OWTs, which are not mapped to the algorithms shown in Table 3, are not assigned a chlorophyll-a or TSI value.

Table 3. Algorithm selection and optimisation strategy for chl-a retrieval per optical water type (OWT)

OWT number	Algorithm source	Algorithm optimisation
1, 5	OC2Ev6 oceancolor.gsfc.nasa.gov/cms/atbd/chlor_a	Empirical re-tuning of algorithm parameters based on UK NERC GloboLakes calibration against the *Limnades* database for MERIS matchups, specific to each optical water type (Neil et al. 2019), applied routinely to OLCI imagery, followed by alignment
2, 9	708/665 empirical band ratio based on Gilerson et al. (2010)	
7	Semi-analytical NIR-Red band algorithm for MERIS based on Gons et al. 2005.	

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		of algorithm responses for MSI compared to coincident products derived from OLCI.

The first group of OWTs (1 and 5) use the Ocean Colour chlorophyll-a algorithm design based on a blue-to-green band ratio. The algorithm parameters were tuned to align the response of MSI-based processing with coincident OLCI products over a wide range of lakes, as detailed in Warren et al. (2021). This alignment is applied to each MSI algorithm in *Calimnos.* The OLCI and MERIS algorithms, in turn, derive their algorithm optimisation from in situ data coincident with MERIS, obtained from the *Limnades* database (https://www.globolakes.stir.ac.uk/limnades.html). This calibration of the downstream algorithms compensates for the fact that the most prominent source of uncertainty in inland water quality retrieval is the atmospheric correction step.

The remaining algorithm groups are variations on empirical and semi-analytical retrieval methods that focus on interpreting the red to near-infrared part of the spectrum. These methods are the most successful approach to retrieve chlorophyll-a concentration over a range of turbidity and trophic levels, because the red part of the spectrum is least influenced by overlapping absorption signatures of dissolved substances and other phytoplankton pigments. However, it should be noted that the Sentinel-2 bands are too wide to capture only the chlorophyll-a absorption feature, which leads to loss of sensitivity compared to results obtained with current ocean colour sensors.

The Gilerson et al. (2010) algorithm is an empirically tuned ratio of bands 708 and 665 nm. The calibrated algorithm is used in *Calimnos* for OWTs 2 and 9.

The Gons et al. (2005) algorithm (for OWT 7) uses the same band ratio and additionally analytically retrieves the backscattering coefficient from the 783 nm band. Subsequently, the absorption at 665 nm is analytically retrieved by inverting the Gordon reflectance model and attributed to chlorophyll-a and water. Empirical tuning is introduced to address some of the sensitivity issues described above.

3.2.7.5 Turbidity (version 2.0 onwards)

Turbidity is obtained from the algorithm described in Nechad et al. (2010, 2015, 2016), using the most suitable waveband in the red or near-infrared part of the spectrum, according to the optical water type. Algorithm mappings to OWTs in *Calimnos* v1.3 are shown in Table 4. Compared to chlorophyll-a, an algorithm is also assigned to OWT 10, with 4 OWTs remaining without being mapped onto any algorithm.

Table 4. TUR algorithm selection and optimisation strategy, per optical water type (OWT)

OWT number	Algorithm source	Algorithm optimisation
1, 2	Nechad et al. (2010, 2015, 2016)	Using 665nm band
5	Nechad et al. (2010, 2015, 2016)	Using 705nm band
9	Nechad et al. (2010, 2015, 2016)	Using 783nm band
7, 10	Nechad et al. (2010, 2015, 2016)	Using 865nm band

3.2.7.6 Floating cyanobacteria and floating macroalgae indices

The spectral bands available on Sentinel-2 are limited for detecting cyanobacteria as some of the spectral bands available on Sentinel-3 are missing. Due to the lack of a waveband at 620nm, which is used within the MPH algorithm (Matthews & Odermatt, 2015) for OLCI, a different methodology is applied to Sentinel-2 data. For the detection of cyanobacteria surface blooms with Sentinel-2 MSI, the Floating Algae Index (FAI) was adopted (Dogliotti et al. 2018, Oyama et al. 2015). Similar to the MPH-based approach, the FAI is based on the principle that the reflectance signal of floating cyanobacteria has unique spectral characteristics that can be distinguished from other types of aquatic vegetation and algae. Differences between the reflectance values of the red and short-wave infrared (SWIR) wavelengths are used and this difference is then normalised by the sum of the reflectance values at these wavelengths, producing a ratio that is used to estimate the presence of floating biomass in the water body. The workflow for Sentinel-2 MSI is shown in Figure 3. FAI is defined as:

$$\boldsymbol{FAI = B7 - (B4 + (B11 - B4) * \frac{783 - 665}{1612 - 665}}$$

Additionally, the Normalised Difference Water Index ($NDWI_{swir}$) based on the Sentinel-2 SWIR (shortwave infrared) bands is used to distinguish between floating cyanobacteria and floating macrophytes:

$$NDWI_{swir} = \frac{B7 - B11}{B7 + B11}$$

The probability for each pixel is subsequently calculated with an exponential function using the $NDWI_{swir}$ and FAI values. This probability indicates the likelihood of floating cyanobacteria or floating macrophytes in each pixel. The weighting coefficients in the probability function are α=1 and β=7.

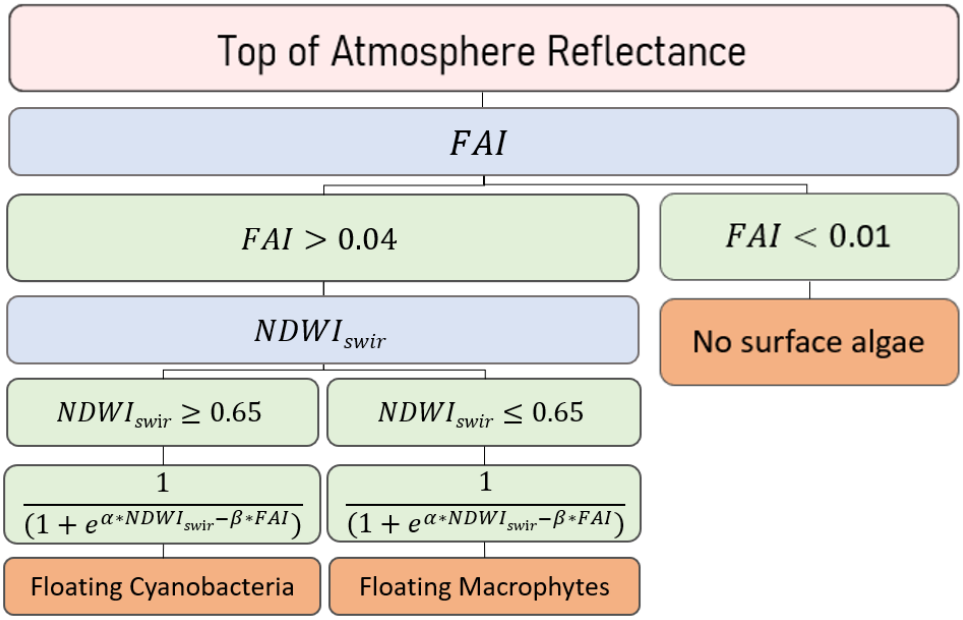


Figure 3. Workflow of detecting surface blooms with Sentinel-2 MSI

The thresholds used in the above scheme have been calibrated using a Sentinel-3 OLCI calibration image as “ground truth” data. Even though the selected EO product for Sentinel-2 MSI was dated three days later, it showed very similar conditions of floating cyanobacteria blooms and floating macrophytes.

3.2.7.7 Temporal Aggregation

The processed satellite overpasses within each 10-day period are aggregated by using the SNAP LakeAggregate operator v5.0.9. It performs spatial and temporal aggregation of pixel values into cells ('bins') of a planetary grid over a lake. During the ‘Lake Aggregation’ step, invalid pixels are eliminated before averaging, using the masking expressions listed in Table 5.

Table 5. Valid pixel expressions

Source	Flag or condition	Applied to
IdePix	INVALID LAND, VEG_RISK SNOW_ICE CLOUD, CLOUD_BUFFER, CLOUD_SHADOW, CLUSTERED_CLOUD_SHADOW,	all variables

Source	Flag or condition	Applied to
	CLOUD_AMBIGUOUS, CLOUD_SURE, CIRRUS_SURE, CIRRUS_AMBIGUOUS	
Polymer	OUT_OF_BOUNDS	all variables
Bright pixels	$R_w443 > 0.4$ or $R_w865 > 0.4$	all variables
Turbidity	Turbidity < 0	turbidity
TSI	TSI = Infinity	trophic_state_index

Both standard deviation ('sigma') and mean values are derived for the remaining observations over a pixel, however, only the mean values are currently included in the output product.

Most representative spectrum selection: for each pixel, fully normalised water-leaving reflectance spectra are compared to the median spectrum using the root-mean-square difference across all wavebands. The spectrum with the lowest RMS difference is selected as most representative for the observation period.

WIC S1+S2 layer: We apply the following criteria for the 10-day (dekad) aggregation, using **number of counts** for each WIC S1+S2 layer value (1, 100, 200, 205, 254, 255 – see Table 6):

If WIC class '100' count in the dekad >= 2, output = 100

Else if WIC class '1' is present (count > 0), output = 1

Else, majority class wins (highest count)

All data are also filtered based on the data producers' quality assessment using WIC_QA class == 0 or 1 (Table 6) so that we have high confidence in the final output that is provided to users.

Table 6. Water and Ice Cover (WIC) S1+S2 product information used in LWQ-100m v2.1 product (Source: HR-WSI PUM 2025)

WIC class	Description	WIC_QA class	Description
1	Open water	**0**	**High quality**
100	Snow-covered or snow-free ice	**1**	**Medium quality**
200	Radar shadow/layover/ foreshortening	2	Low quality
205	Cloud or cloud shadow	3	Minimal quality
254	Other features (e.g., land)	200	Radar shadow/layover/ foreshortening

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WIC class	Description	WIC_QA class	Description
255	No data	205	Cloud or cloud shadow
		255	No data

3.2.8 Limitations

MSI, in contrast to ocean colour sensors, has relatively broad and fewer wavebands. It should be expected that, even when the bands are centred on diagnostic absorption features of optically active water constituents, sensitivity of the sensor to quantify these substances will be limited compared to ocean colour sensors. This can manifest in the products as having fewer valid outputs (more masked pixels), and result in showing a higher degree of uncertainty by comparison to coincident observations with ocean colour sensors.

The major limiting factor in (i) assessing the accuracy of water-leaving reflectance, (ii) the OWT-specific mapping of water constituent concentrations, and (iii) the accuracy of the water-column products is the scarcity of available in situ data to support algorithm tuning and validation. Without dedicated field campaigns, automated monitoring stations, and community data sharing arrangements, this remains a major source of product uncertainty. On the other hand, the waveband selection on the MSI sensor approximates that of key wavebands on ocean colour sensors, including bands located around the ‘red edge’ of the spectrum, which is relevant to observing phytoplankton in optically complex water types.

Comparison of the Sentinel-2 derived products against better established algorithms for MERIS and OLCI has resulted in the first set of calibrated algorithms via alignment of MSI to OLCI. Continued analysis against a growing volume of in situ observations is required to independently optimise MSI-based estimates, which may resolve gaps in the assignments of the algorithm selection to several OWTs.

Some uncertainty remains regarding the treatment of observations within the overlap of adjacent MSI detectors during mapping to L1C. Without accurate unmapped (L1B) data, there is a risk that averaging of observations obtained under different viewing angles is not accurately handled at the atmospheric correction stage leading to visible camera boundaries in derived products.

The 100m products include near-shore areas that may be shallow. Such areas are not yet flagged. One method that was investigated (ESA Diversity-II project, 2015) consists of a long-term analysis to identify suspected shallow areas, but this method has not been globally verified.

In addition, it is tempting to assume that a higher resolution satellite sensor will successfully resolve water quality in smaller water bodies. This is true only to the extent where pixels are relatively unaffected by reflected light from adjacent land, mixed with the water leaving reflectance in a turbid atmosphere. Best results may be expected under optically thin atmospheric conditions, and in water bodies where the contrast between water and nearby land is limited. These conditions

are most likely encountered in turbid water bodies and at locations where vegetation is a minor land cover class.

Finally, the use of the LAND flag from *IdePix* is based on a static high-resolution land mask. There are known (usually small) water bodies which were dry during the period investigated to produce this land mask. As a result, these water bodies do not show valid results even during periods when they are filled. To improve on this method, a dynamic land/water detection would need to be globally tested and implemented.

3.3 QUALITY ASSESSMENT

3.3.1 Quality Flags

For the biogeophysical variables, each pixel in the final products is an aggregation of many observations. The number of underlying observations will vary as some inputs are masked due to insufficient quality, contamination by cloud or ice and other factors. The *num_obs* variable gives the total number of input observations to a pixel before quality control. A *quality_flag* is provided which indicates what masking has removed input observations from the aggregated final output. Flags raised in this variable do not indicate a degradation in the quality of the final aggregated pixel, but a reduction in the number of input observations which may reduce the representativeness of the result over the aggregation period. Table 7 shows the meaning of these quality flags.

Table 7. Quality flag definitions

Flag name	Definition	Explanation
bright_pixels	Rw412 > 0.4 or Rw865 > 0.4	One or more input observations removed as anomalously bright
cloud	pixel_classif_flags.IDEPIX_CLOUD or pixel_classif_flags.IDEPIX_CLOUD_BUFFER or pixel_classif_flags.IDEPIX_CLOUD_SHADOW or pixel_classif_flags.IDEPIX_CLOUD_AMBIGUOUS or pixel_classif_flags.IDEPIX_CLOUD_SURE	One or more input observations removed as were flagged by *Idepix* as being affected by cloud
snow_ice	pixel_classif_flags.SNOW_ICE	One or more input observations removed as flagged by *Idepix* as snow or ice

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Flag name	Definition	Explanation
l1_invalid	l1_flags.invalid	One or more input observations removed as instrument data missing or invalid, flagged in source data
land	land_water_fraction < 100	One or more input observations removed as some percentage of the pixel is land
land_contaminated	(owt_14 + owt_15>1.59) and ((owt_dominant_class==14) or (owt_dominant_class==15))	One or more input observations likely contaminated by adjacency effect, but not removed (source variables for this expression not supplied)
atmospheric_correction_failure	Polymer bitmask: OUT_OF_BOUNDS	One or more input observations removed as Polymer bio-optical model reached its concentration limits
turbidity_invalid	turbidity > 100000 or turbidity < 0	One or more input observations removed as turbidity value likely unrealistic (suspected unmasked ice)
chla_invalid	Chla < 0 or Chla > 10000	One or more input observations removed as chl-a value likely unrealistic
tsm_invalid	TSM < 0 or TSM > 10000	One or more input observations removed as TSM value likely unrealistic
floating_cyanobacteria	floating_cyanobacteria > 0.4 and not (l1_invalid, cloud, land, land_contaminated)	Floating cyanobacteria likely in one or more input observations

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Flag name	Definition	Explanation
floating_macrophytes_or_mixed_pixel	floating_macrophyte_probability (not supplied) > 0.5 and not (l1_invalid, cloud, land, land_contaminated)	One or more input observations likely contained floating macrophytes or was a mixed pixel

3.3.1.1 Cloud screening

IdePix is used to generate masks for Sentinel-2 data. Visual, qualitative analysis of the performance of this processing stage has been previously carried out, resulting in the combination of masking criteria currently used in the production chain.

The CLOUD flag consists of different elements, dedicated to different cloud types: CLOUD_SURE, CLOUD_AMBIGUOUS, CLOUD_BUFFER, CIRRUS_SURE and CIRRUS_AMBIGUOUS. Currently, the CLOUD_SURE, CLOUD_BUFFER and CLOUD_AMBIGUOUS flags are applied, and no values are consequently provided for affected pixels. The visual tests indicate that it is necessary to further apply the CLOUD_SURE, CLOUD_AMBIGUOUS and CLOUD_BUFFER flags, as well as the CIRRUS_SURE and CIRRUS_AMBIGUOUS flags. Figure 4 shows examples of the different cloud flagging configurations. Figure 5 illustrates how increasingly progressive cloud screening behaves in relation to turbidity retrieval for Lake Garda, while Figure 6 shows the cloud products in an RGB scene of Lake Turkana. Artefacts are visible in the TUR products where cloud screening was insufficient for this specific application. Adding the CIRRUS flags improved this result.

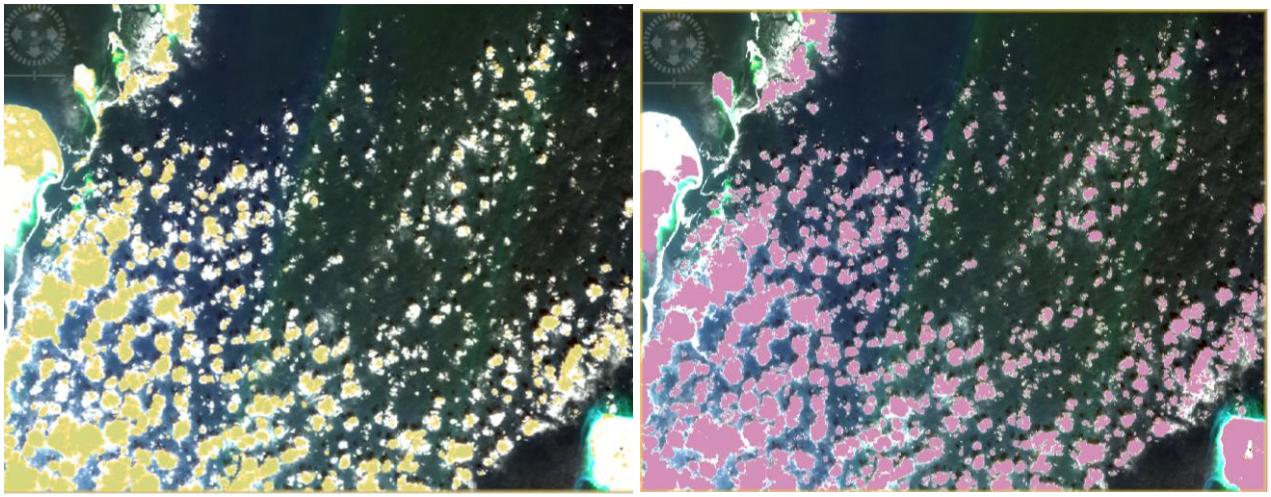


Figure 4. Cloud flag on small and thin cumulus clouds in Lake Michigan (22.03.2017). Left: cloud and CLOUD BUFFER (2pixels); right: CLOUD and CIRRUS_SURE flag

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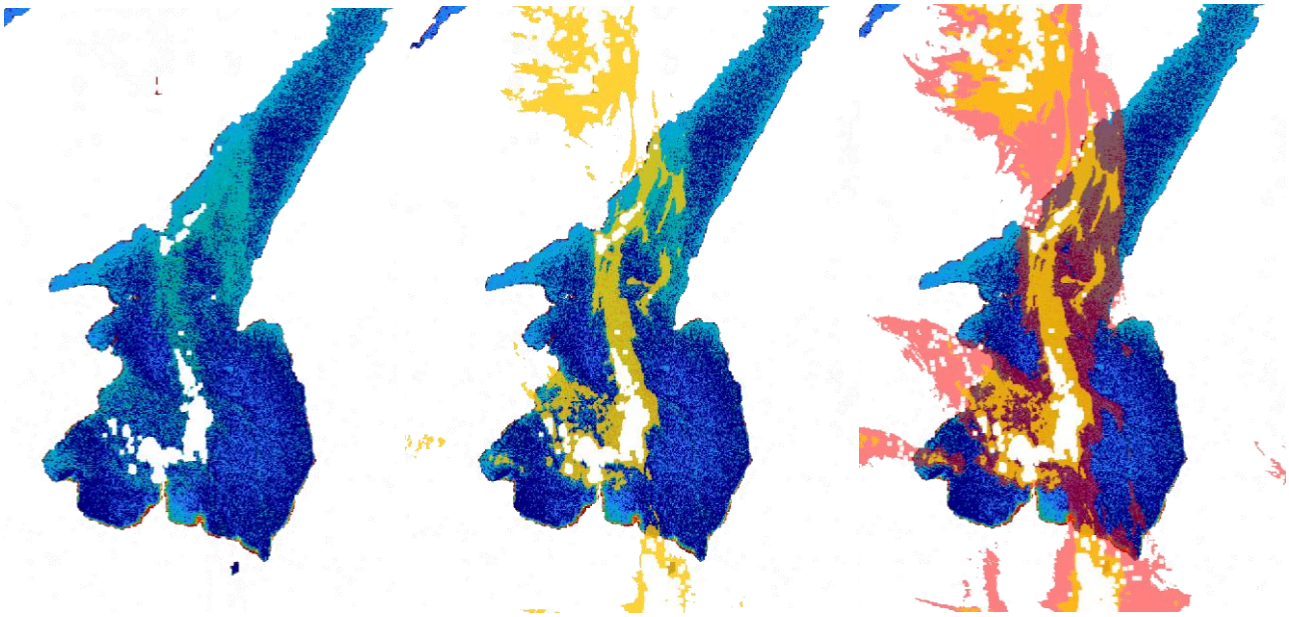


Figure 5. Blended TUR in Lake Garda: left: all overlaid with CLOUD, CLOUD_BUFFER and CLOUD_AMBIGIOUS FLAGs; middle: in addition, the CIRRUS_SURE flag is raised; right: in addition the CIRRUS_AMBIGIUOUS flag is raised

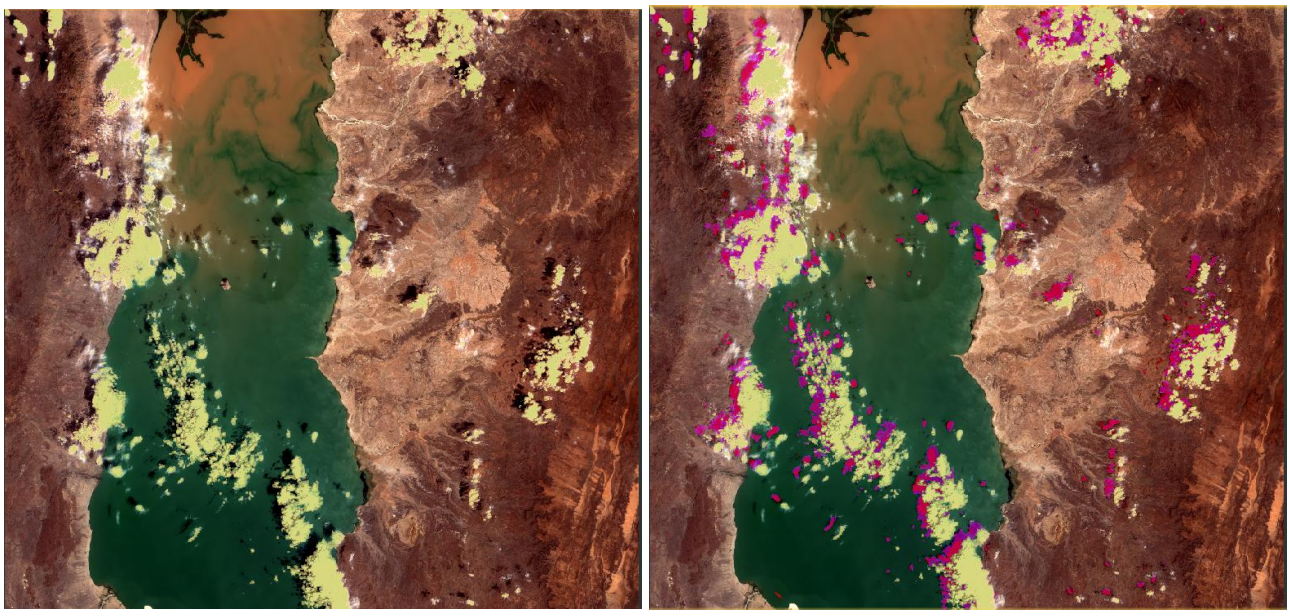


Figure 6. Lake Turkana (Kenya, Ethiopia) from 2018-01-27, RGB, with *IdePix* cloud mask flagging CLOUD and CLOUD_BUFFER (yellow) (left) and cloud shadow flag (pink) (right)

3.3.1.2 Shoreline detection

The standard (static) land flag for Sentinel-2 is not adequate for inland waters. Shoreline pixels remain and are included in the water mask (Figure 7). To mitigate this, the *IdePix* VEG_RISK flag

is used to identify the land-water boundary and flag mixed pixels of land and water. Therefore, these pixels are excluded from the final product.

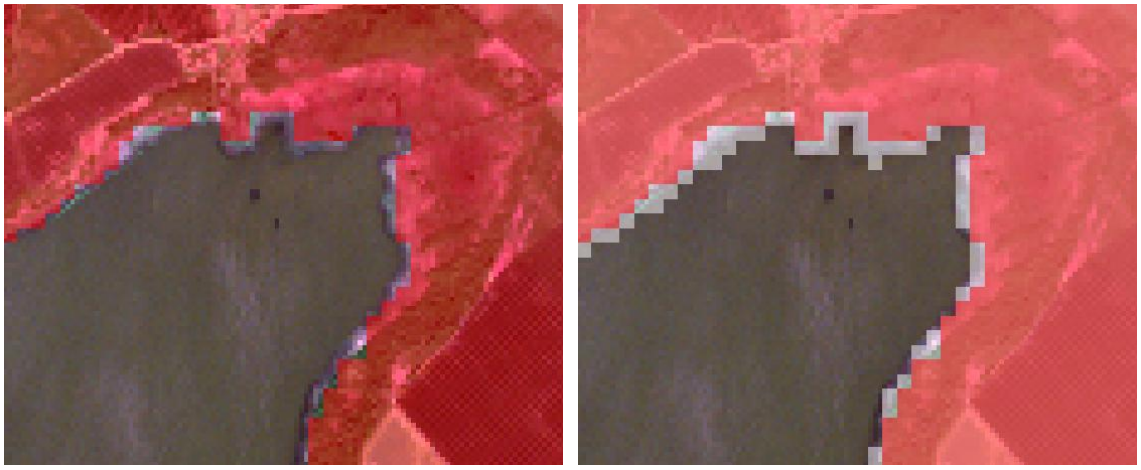


Figure 7. RGB image overlaid with IDEPIX_LAND (red) and on the right also with IDEPIX_VEG_RISK (white) flags. The VEG_RISK flag identifies the mixed pixels at the shoreline. The RGB image has 20m resolution; flags are overlaid in 60m resolution to show mixed pixels within 60m products

A second flag to detect shoreline pixels is the poly_OUT_OF_BOUNDS flag which is raised by Polymer when the optimised parameters of the bio-optical model (log chlorophyll-a and f_b) depart from within a predefined range. Comparison between the IDEPIX_VEG_RISK and poly_OUT_OF_BOUNDS flags shows that the latter identifies more shoreline pixels but also flags some shallow waters (Figure 8).

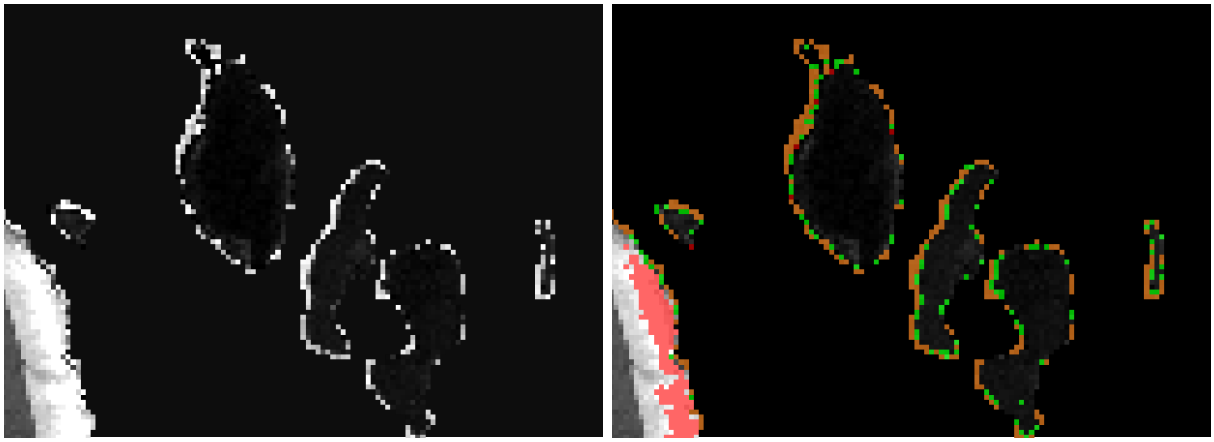


Figure 8. (left) Shoreline pixels (mixed pixels) not flagged by LAND flag show higher values in R_w560. (right) Flagged with poly_OUT_OF_BOUNDS (red) and *IdePix* (green)

3.3.1.3 Flags for floating cyanobacteria and floating macrophytes

Visual inspections and anecdotal records from articles and social media posts have been used to identify occurrences of and assess the ability of the satellite products to represent surface blooms.

Several CyanoTracker (https://twitter.com/CyanoTracker) posts included blooms that were visually identified from satellite images across various lakes. These posts were used to locate bloom events and test the methodology described here. Similarly, a range of published media articles related to these bloom events were studied to specify the type of bloom that took place. It was found that all events could be consistently identified. Nevertheless, some uncertainty remains regarding the accuracy of mapping the extent for each floating biomass class. To assess the sensitivity of the class assignments, in situ measurements would be needed for further validation.

Inspection of the spectral shape of identified pixels has been another way to verify the suitability of the two masks.

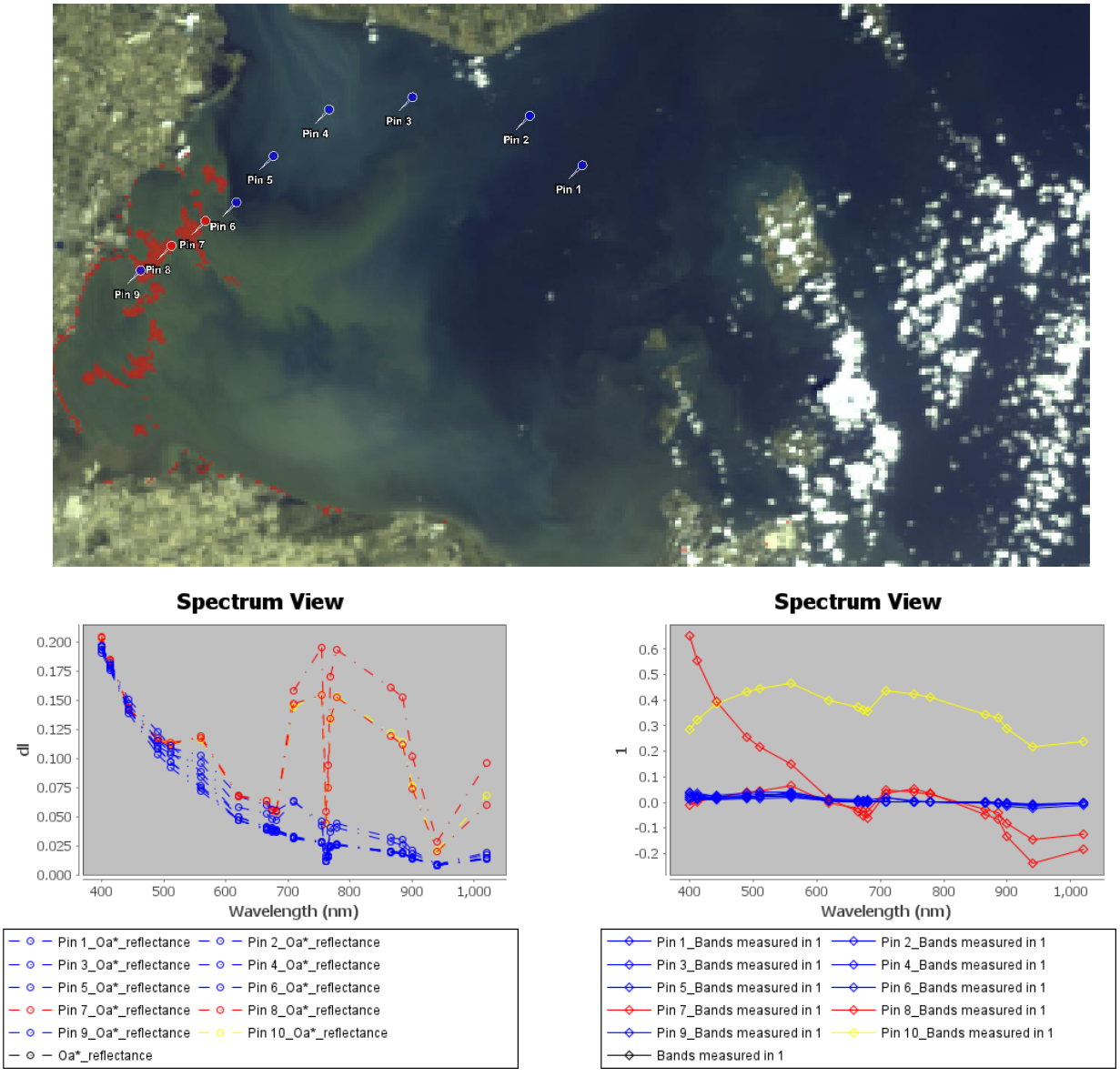


Figure 9. Floating cyanobacteria in western Lake Erie. (top) RGB image from TOA reflectance with floating cyanobacteria mask overlaid in red, and the locations for which spectra are plotted in the

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bottom graphs. (bottom-left) TOA reflectance spectra, (bottom-right) atmospherically corrected spectra

3.3.2 Algorithm performance

3.3.2.1 Atmospheric correction algorithm selection

An assessment of six atmospheric correction algorithms was performed, with details of the analysis given in Warren et al. (2019). The output of each algorithm was validated against in situ reflectance from inland water bodies as well as in situ reflectance from coastal regions (Baltic Sea and Western Channel UK). Coastal data were added to the dataset to provide a wider range of observations because lake observations coinciding with satellite imagery are still scarce. The resulting dataset is described in Figure 10 showing the results for Polymer v4.6 (an earlier version of Polymer, released 23.11.2017) Sentinel-2A AC at 60m resolution against in situ reflectance (both shown as remote-sensing Reflectance, R_{rs}, with no BRDF correction applied to in situ data and a factor π used to derive R_{rs} from Polymer R_w) for each band of the Polymer output. Table 8 shows the distribution of in situ data across the different water bodies and the number of Sentinel-2 granules for which comparative data were obtained. Figure 11 shows a plot of the coefficient of determination, root mean square (RMSE), mean absolute percentage difference (MAPD) and bias for Polymer v4.6 separated by region.

Table 8. Number of MSI granules with valid matchups, after removing duplicates, in the time windows of ±3 h for Western Channel, UK, and Baltic Sea datasets and ±24 h for the inland dataset

	Inland	Baltic	Western Channel
granules	35	30	5
matchups	230	986	452

Warren et al. (2019) found that Polymer was one of the best performing algorithms for MSI when all statistical metrics and the number of valid retrievals were considered. However, retrieval of R_{rs} shows a high degree of uncertainty and under-estimation in several bands that are essential for the accurate retrieval of chlorophyll-*a* on which the TSI product is based. Retrieval of R_{rs} at coastal sites was generally better than for inland waters. It is not unlikely that this reflects the negative influence of nearby land (‘adjacency effect’) on retrieval accuracy which is particularly problematic in small water bodies.

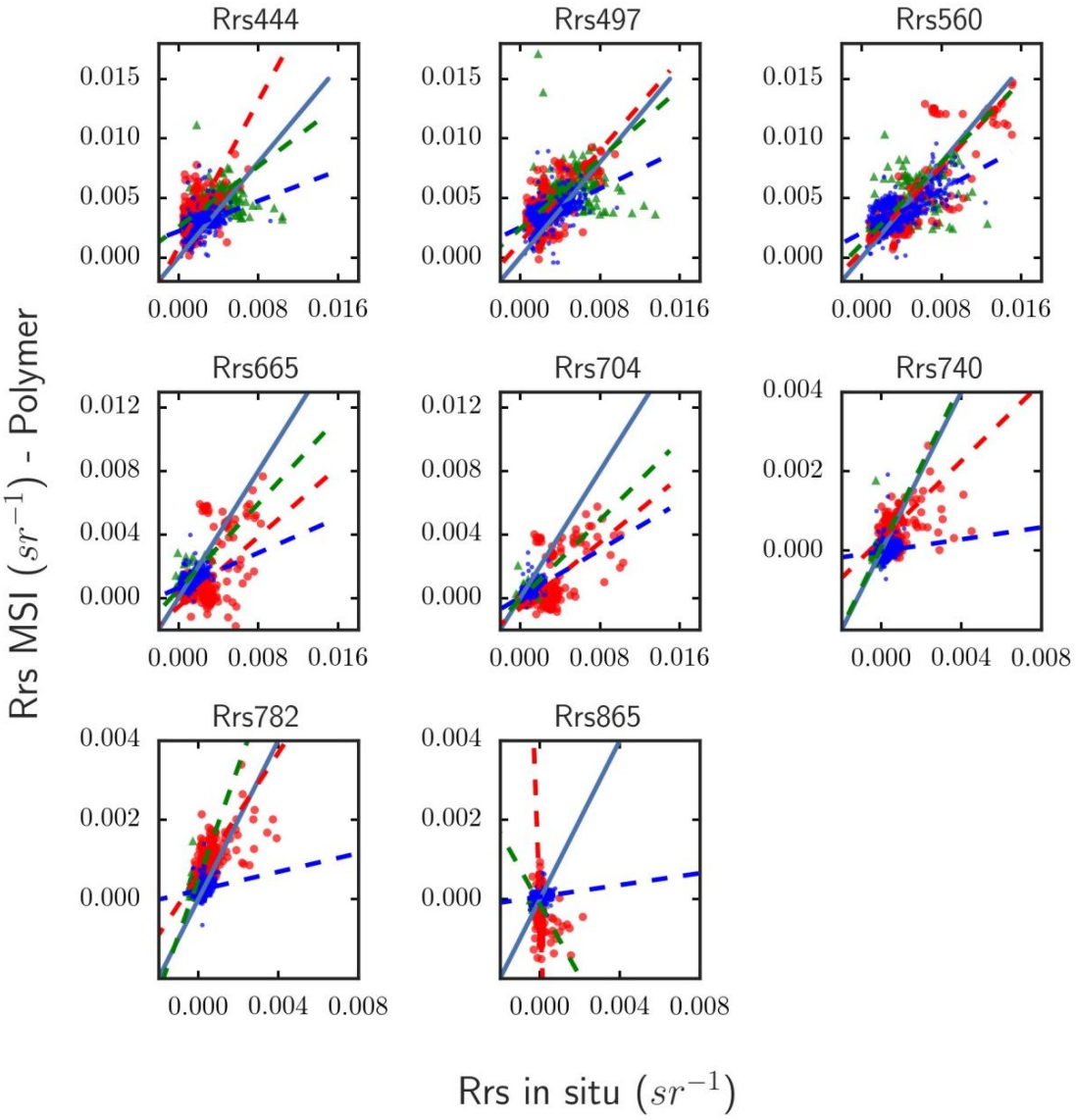


Figure 10. Polymer v4.6 plotted as a function of in situ reflectance (both as remote-sensing reflectance, R_{rs}, units sr^{-1}) for each corrected MSI band. There are 1735 matchup points: Baltic Sea (blue), Western Channel (green) and inland waters (red). Solid lines mark unity whereas dashed lines are orthogonal distance regression fits for the respective regions (see uncertainty metrics in Figure 11) (Source: Warren et al. 2019)

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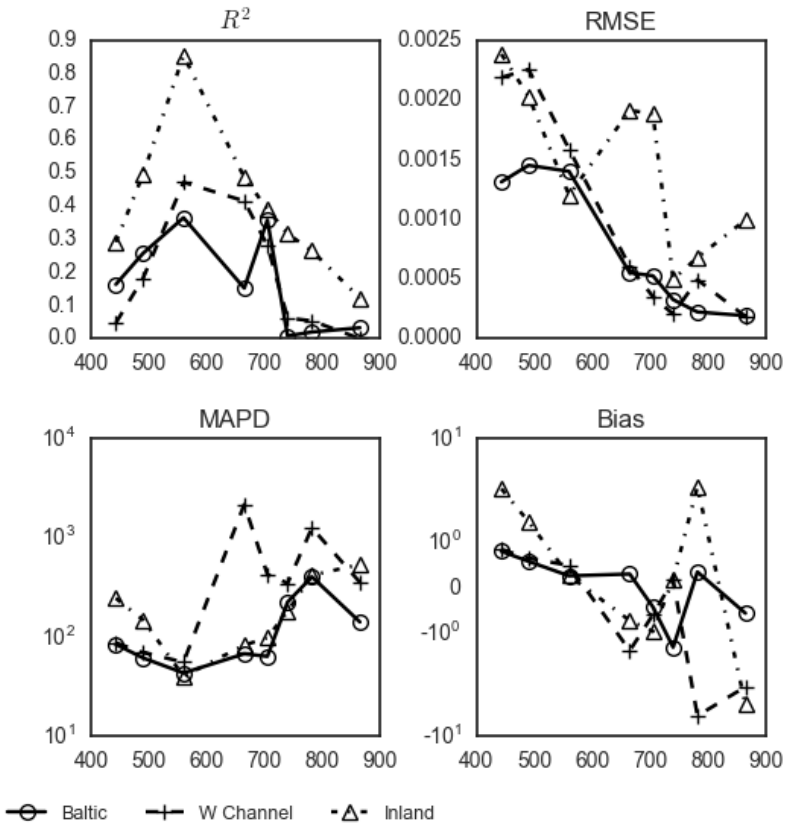


Figure 11. Statistics of Polymer matchups plotted for each region (circles: Baltic Sea, plusses: Western Channel, triangles: inland waters). The metrics are (from top left): coefficient of determination (R2), root mean square error (RMSE), mean absolute percentage difference (MAPD) and bias

3.3.2.2 Atmospheric correction algorithm performance vs in situ

The performance of Polymer v4.14, as used in product version v2.0, was assessed with comparisons against in situ R_{rs} from the GLORIA inland water reflectance dataset (Lehmann, 2023, Figure 12) and the WISPCloud dataset (Figure 13). R_{rs} for MSI from this Polymer version generally performs well in comparison with in situ R_{rs}. A slight negative bias is seen *vs*. the WISPCloud dataset, and R_{rs} for Lake Eire is underestimated when compared with the GLORIA dataset.

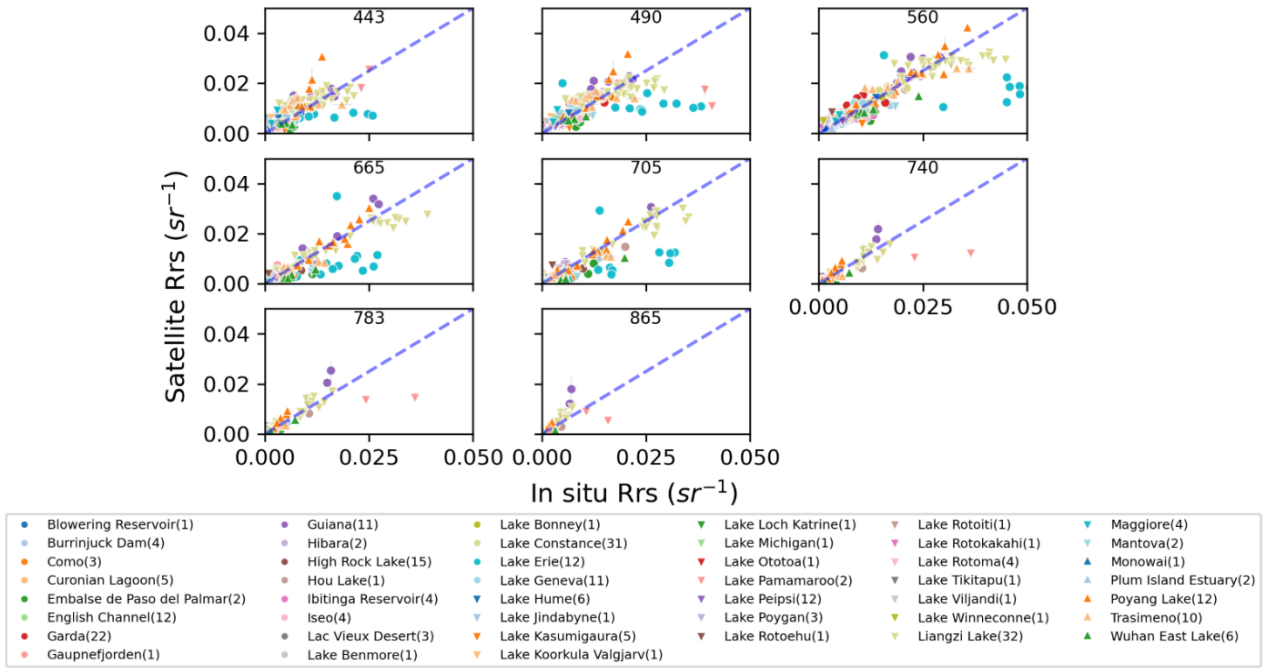


Figure 12. Matchups of in situ R_{rs} from the GLORIA inland water dataset and Sentinel-2 MSI R_{rs} derived using Polymer v4.14

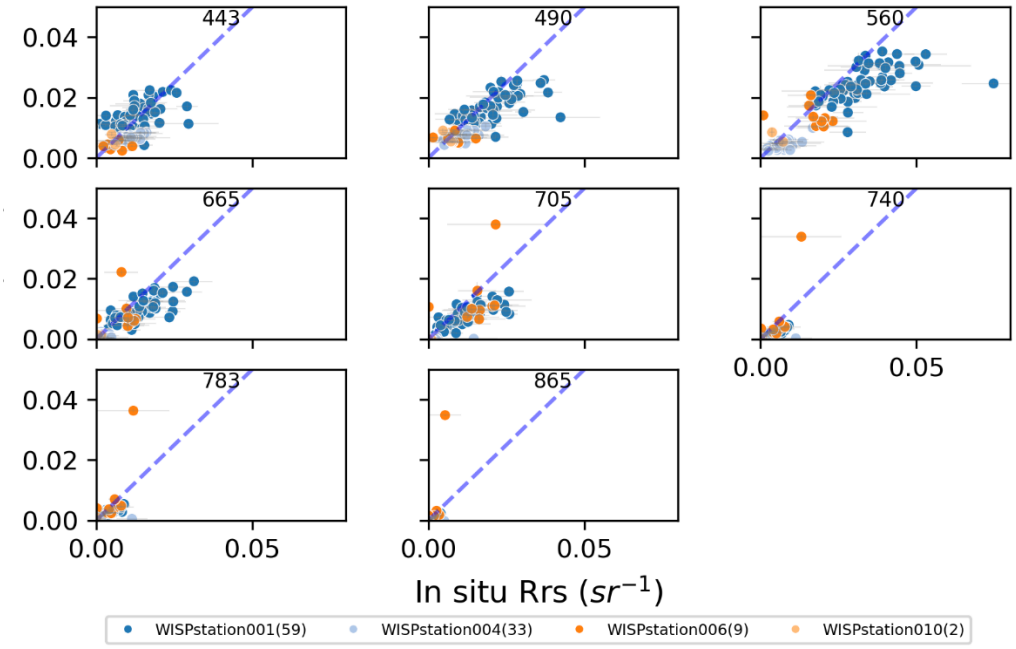


Figure 13. Matchups of in situ R_{rs} from WISPCloud and Sentinel-2 MSI R_{rs} derived from Polymer v4.14

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3.3.2.3 Spatio-temporal consistency between sensors

For the investigation of consistency between sensors, time series plots have been generated that show the seasonal trends of 2 years of Sentinel-3 OLCI 300m (v1.4) compared to Sentinel-2 MSI 100m products (v1.5). Figure 14 shows the time series at one virtual station in Lake Mueritz for each variable (water leaving reflectance, turbidity and TSI). The OLCI 300m time series extends from January 2020 to April 2022 (marked with crosses), while 100m products are available from July 2021 to April 2022 (circles).

The example shows that for the water leaving reflectance spectra (443 to 865nm) the Sentinel-2 data fits well within the range of the corresponding OLCI data, though R_w665 and R_w705 are a bit higher than OLCI values. Turbidity is in the same range (below 2FNU) in both sensors, but MSI turbidity is substantially lower in most of the cases. Then again, TSI from MSI is in good correspondence with OLCI.

Further examples are provided in the QAR. Overall agreement is observed between the sensors, with exceptions for highly turbid waters (Lake Balaton) and underestimation of turbidity in clear waters (Lake Garda).

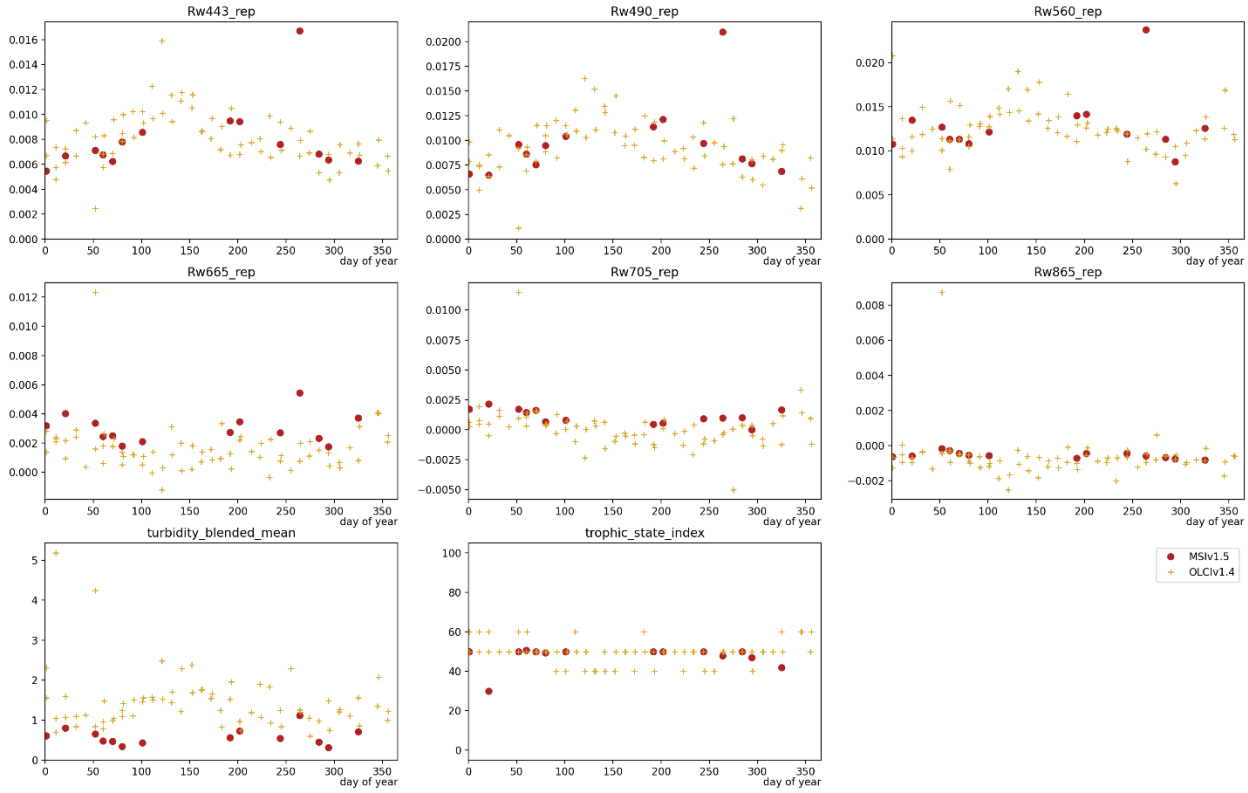


Figure 14. Sentinel-3 OLCI 300m and Sentinel-2 MSI 100m seasonal comparison for central Lake Mueritz

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Consistency has further been tested in the spatial domain (by comparison of values along transects, examples shown in Figure 15) and in the spectral domain by comparison of reflectance spectra for same time acquisitions. More examples are provided in the QAR.

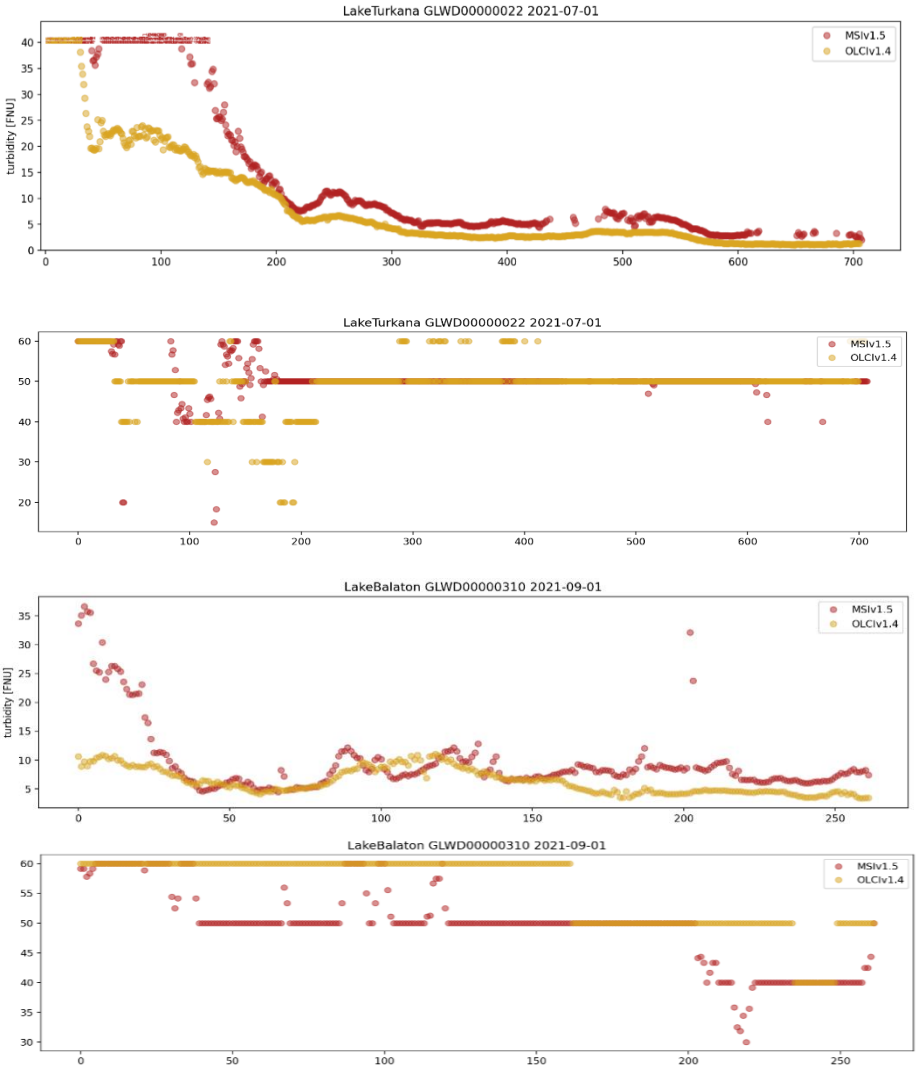


Figure 15. Transect plots for Sentinel-3 OLCI 300m products and Sentinel-2 MSI 100m products for Lake Turkana and Lake Balaton

Transect plots include both examples of agreement and disagreement between the sensors. While there is generally good agreement between OLCI and MSI, turbidity is higher for MSI than for OLCI in the most turbid regions of these two example lakes.

3.3.2.4 Floating cyanobacteria and macrophytes

For the validation of floating cyanobacteria, reports available via online platforms or social media of blooming events were used. Reports for macrophytes (sargassum, hyacinths) as well as cyanobacteria were used in order to verify that the differentiation works properly.

Using observations from CyanoTracker and media articles, all events could be consistently identified. Nevertheless, some uncertainty remains regarding the accuracy of mapping the extent for each floating biomass class. To assess the sensitivity of the class assignments, in situ measurements would be needed for further validation.

In a second step, the spectral information within the detected bloom and in the surrounding water were compared and assessed. We assume that the water around surface blooms of cyanobacteria is affected by immersed cyanobacteria.

The image from Winam Gulf shown in Figure 16 highlights areas identified as having cyanobacteria (predominantly in the southwest) and macrophytes (in the centre of the image and further east). Spectra in the open water around these occurrences (plotted in blue) do not show distinct differences in shape, noting that phycocyanin detection is not possible due to absence of a band in the 620nm region. Spectra in the floating macrophyte mask show a sharper rise in the 740-780nm region but are otherwise similar to those associated with floating cyanobacteria. Compared to macrophytes, the cyanobacterial surface blooms show a higher absorption in the SWIR bands, indicating that the effects of water absorption on cyanobacterial blooms are stronger than those on macrophytes. Since the water absorption in the SWIR band is much stronger than that in the NIR bands, the SWIR band is considered more effective than the NIR band for distinguishing cyanobacterial blooms and aquatic macrophytes.

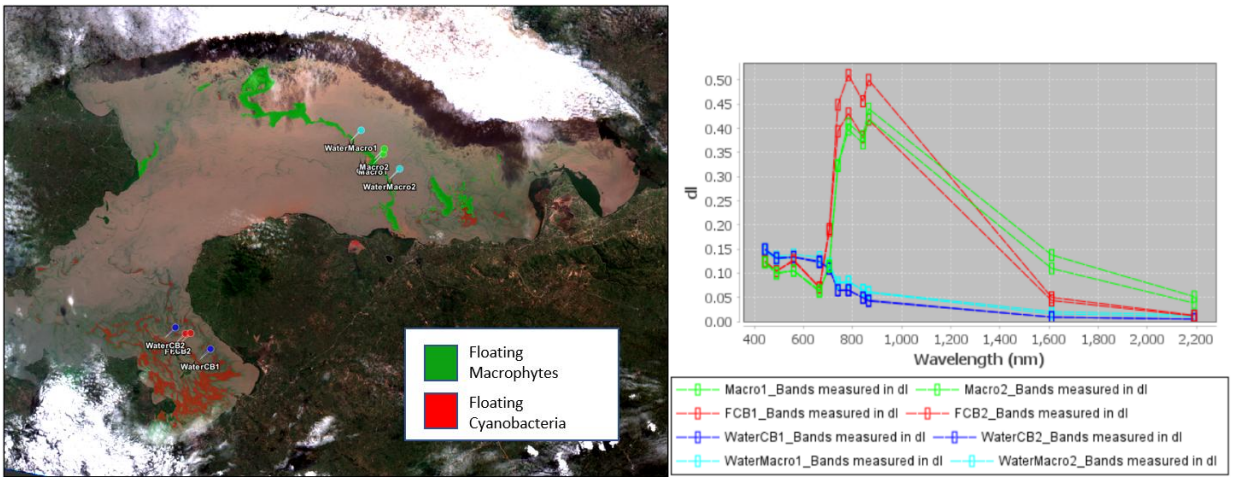


Figure 16. Spectra associated with areas identified as having floating cyanobacteria and macrophytes in Winam Gulf, Lake Victoria. Key to the spectra: Blue – open water in vicinity of floating cyanobacteria, Red – floating cyanobacteria identified, Light Blue – open water near macrophytes, Green – floating macrophytes

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3.3.3 Discontinuities in highly turbid water

Visual artifacts have been observed in the most turbid lakes in the dataset, e.g. Lake Tana (Figure 17). By applying the poly_OUT_OF_BOUNDS flag we can remove the regions which show spatial discontinuities. Whilst this is effective, it results in data loss (Figure 17) and may introduce bias to the results towards favouring clearer waters, which is undesirable.

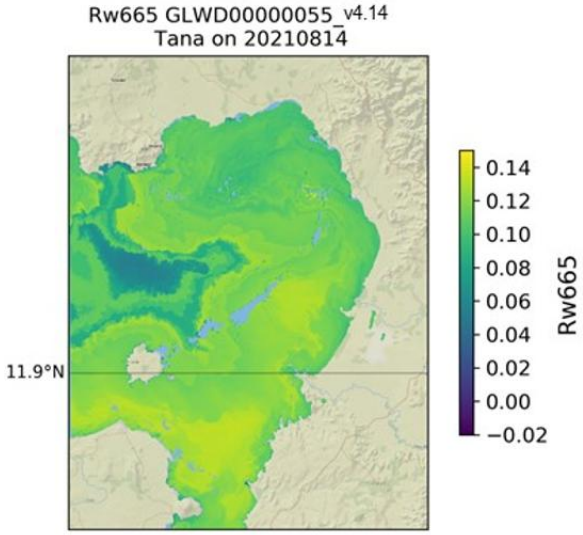


Figure 17. Sentinel-2 MSI water leaving reflectance at 665nm from Polymer v4.14 showing artifacts observed in highly turbid water

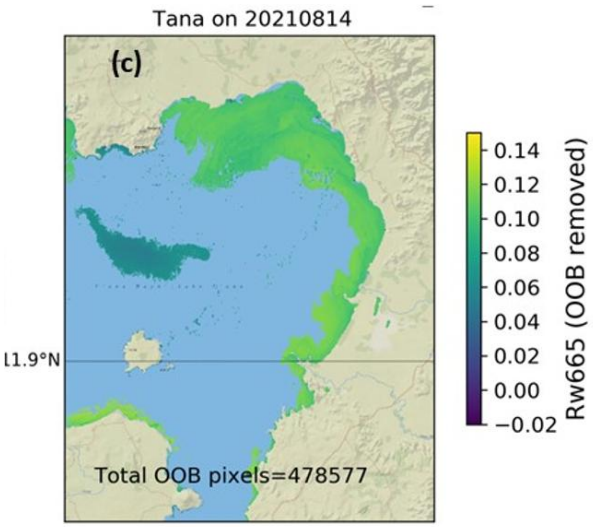


Figure 18. Artifacts removed using the poly_OUT_OF_BOUNDS flag but with significant loss of data

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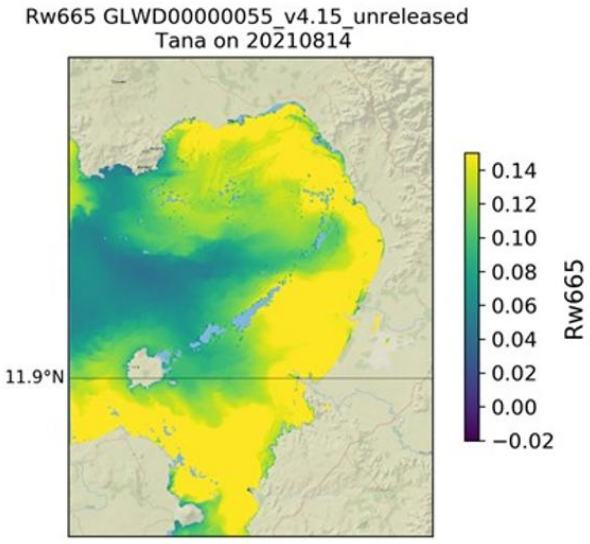


Figure 19. Water leaving reflectance at 665nm from Polymer v4.15

A new version of Polymer (versions 4.15 and higher) has been released in response to this issue. A range of initialisation conditions are tested for each pixel, and the mineral absorption model is applied based on a log(chl) threshold rather than being permanently on or off. The new Polymer versions (v4.15 and higher) are effective in tackling the high turbidity artifacts (Figure 19) and are, thus, considered the candidate atmospheric correction algorithms for future product v3.0.

Figure 20 and Figure 21 show that for turbid lakes the distribution of reflectance is quite different between Polymer versions 4.14 and 4.15, with v4.15 able to reach higher reflectance at longer wavelengths. The histograms for v4.14 and v4.15 in clear lakes remain similar.

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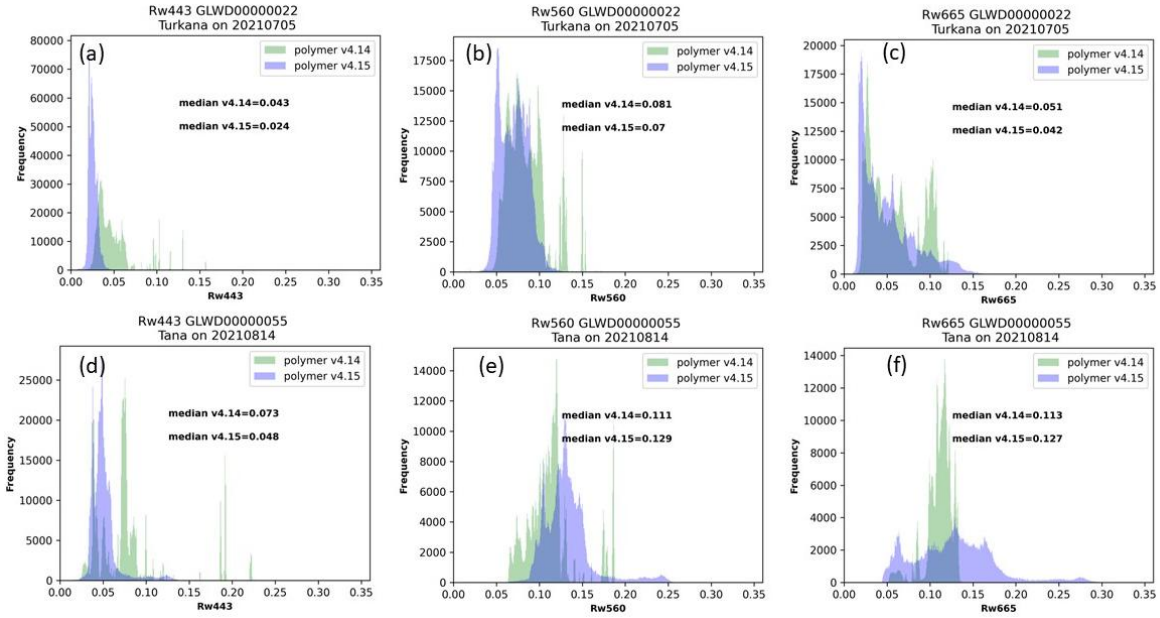


Figure 20. Histogram distributions of R_w Polymer outputs (v4.14 and v4.15) for (a-c) turbid lakes Turkana and (d-f) Tana, at bands 443, 560 and 665nm

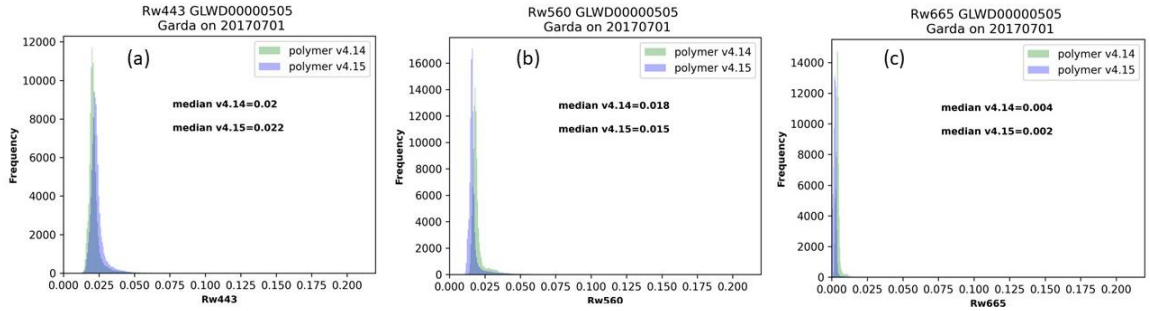


Figure 21. Histogram distributions of R_w Polymer outputs (v4.14 and v4.15) for clear lake Garda, at bands (a) 443nm, (b) 560nm and (c) 665nm

3.4 RISK OF FAILURE AND MITIGATION MEASURES

Existing tools and software have been extended to provide support for handling Sentinel-2 MSI inputs where needed. In general, these modifications have had no impact on the performance of the processing chain for MSI and other sensors. The risk of software failure is, therefore, low.

Should future evaluation prove that modifications to the algorithms are necessary, such changes can be implemented with relative ease. The current processing time for a single MSI granule at 100m resolution is in the order of 20-30 minutes on an average processing node. To limit the processing time, tests and product evaluation are typically performed against subset scenes, thus limiting the risk of delays due to reprocessing efforts.

For the production of LWQ-100m products, *Calimnos* is operated on CDSE. Potential unavailability or instability of CDSE services may cause delays in NRT product delivery. Automated checks and failure notices to the production team are in place to ensure close monitoring, striving to notify users the earliest possible.

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